Searching voices: Towards a canon for interaction design

SEARCHING VOICES

Towards a canon for interaction design

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SCHOOL OF ARTS AND COMMUNICATION | MALMÖ UNIVERSITY | 2003

Studies in Arts and Communication #01

ISSN 1652–0343 ISBN **91–7104–010–2**

Published by the School of Arts and Communication, Malmö University, Sweden.

Series editors: Pelle Ehn, Inger Lindstedt,
Jonas Löwgren, Bo Reimer, Carl Henrik Svenstedt.

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Typeset in Sabon and Futura LT by Jonas Löwgren.

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INTRODUCTION

Pelle Ehn and Jonas Löwgren

In recent years, information and communication technology has taken on whole new meanings in Western society and everyday life. Computers were once productivity tools for industry and administration, clearly alien to the personal life-worlds of most people. The goal of so-called user-centered approaches was to find digital ways of supporting work and enterprise goals more efficiently.

But then something happened. Electronic mail and web surfing became everyday household activities. Computer games grew in popularity to become one of the major sectors in commercial entertainment. Mobile digital services are transforming the way we communicate, share, exist together. Ubiquitous computing is gradually infusing a digital infrastructure into our personal spaces.

In this situation interaction design is emerging as a new and challenging design discipline. This discipline has a design-oriented focus on human interaction and communication mediated by artefacts. The identity and actuality of this emerging design discipline is emphasized by the convergence of digital artefacts with physical space and the objects surrounding us (ubiquitous computing) as well as the convergence between different media (multimedia, new media).

Interaction design has grown out of a merge between the traditional design fields—mainly product design and graphic design—and socially oriented computer studies. The academic heritage in the latter category includes human-computer interaction in particular, but also computer-supported cooperative work and participatory design. Other important initial contributions come from architecture, art, sociology and media studies. *Bringing Design to Software*, edited by Terry Winograd and with contribution from many design fields as well as from computer science was an early manifestation of this development. Design of computer arte-

facts is here understood as an activity that is conscious, is a conversation with the material, is creative, is communication, and is a social activity that keeps human concerns in the center and has social consequences (Winograd, 1996).

However, a question that may be raised about new media in practice is how new the design that is brought in really is. What we meet is in many ways closely related to the tradition from the Bauhaus. As Lev Manovich observes in *The Language of New Media* (2001) we clearly see the traces of »new vision« (Moholy-Nagy), »new typography« (Tschichold) and »new architecture« (le Corbusier).

The real challenge in interaction design is maybe the other way around. The really new is bringing computational technology to design and dealing with ubiquitous computing (Weiser, 1991). The design materials for ubiquitous computing and the appearances of computational artifacts are both spatial and temporal (Hallnäs and Redström, 2002). With computational technology we can build temporal structures and behavior. However, to design these temporal structures into interactive artefacts, almost any material can be of use in the spatial configuration. Interaction design deals with a new kind of combined interactive narrative and architecture.

There are yet no universal definitions, best practices or standardized university curricula, but the emerging interaction design community seems to be built around a few points of common agreement. First, in order to understand what makes a vision of interplay between people and digital technology »better«, it is essential to take an eclectic approach. Digital technology is highly malleable: it has roles to play in most aspects of everyday life. Situations of interplay invariably entail multiple stakeholder perspectives and conflicting intentions. Hence, all of the following statements are true in spite of the apparent contradictions.

- Interaction design supports rather than counteracts existing human practices.
- Interaction design questions existing practices.
- Interaction design creates potentials for action with room for unanticipated actions.
- Interaction design shapes the experience of new spaces in the borderlands between physical and virtual.
- Interaction design creates a sense of awe, inspiring thoughts beyond the apparent.

Secondly, the work involved in interaction design is constructive and intentional. It is about creating artefacts that did not exist before, statements concerning the viability of possible futures. It is best understood as a design discipline. This has quite strong implications for teaching and research as well as for professional practice. It also means that more mature design disciplines such as architecture, product

design and visual communication are valuable sources of knowledge, as witnessed by the interdisciplinary field of design studies.

Finally, the digital material has certain qualities that distinguish it from other design materials. Although highly malleable, it is not entirely shapeless. Its perhaps most important material qualities are the synthesis of spatial and temporal aspects, and the role as a communication medium encompassing »interactivity« as well as many-to-many communication.

The emerging picture of interaction design is one of a rather small core around the peculiar qualities of the digital material, in constant exchange with many other fields of culture, society, technology, science, design and arts. The School of Arts and communication at Malmö University is no exception in this respect. One element of our work in interaction design, along with education on bachelor and master levels, research and artistic practice, is a PhD study program.

The program opens up for a design orientation rather than a traditional PhD approach. Examples indicating this orientation are that:

- the program in general is oriented towards coaching, learning by doing and reflection-on-action;
- focus is on a design-oriented synthesis of constructive action and critical reflection and on synthesis of art and technology;
- students have a commitment to design studies but come from different disciplinary backgrounds;
- research work is carried out in a production-oriented, studio-based interdisciplinary and cross-art environment;
- the thesis may take the form of a portfolio of works and a reflective articulation;
- form is allowed to follow content, hence the form of the thesis may be an interactive multimedia production.

Such a design orientation of the discipline challenges the role traditionally expected to be played in academic life by theories as explicit, abstract, universal and context-independent carriers of knowledge and draws attention to the practice of knowing, to politics, sensuousness, embodiment and particularity.

The kind of support to expect from »design theory« is not so much in terms of »scientific theories« for prediction of results of an activity independent of its context and situation, but rather support for reflections about conditions for changed human activity. Such »design theories« are rather practical instruments to support the designer as a reflective practitioner (Schön, 1987) to improve his or her compe-

tence or design ability to make ethic and aesthetic judgments that are appropriate in their contexts.

The following collection of papers illustrates one element in the formation of interaction design as a field of knowledge, namely the eclectic exchange with other fields through appropriation and adaptation. The PhD students draw on canonical work in their native fields in order to understand what »better« means in their respective views of interaction design.

The PhD students represent a broad variety of academic and artistic backgrounds, including computer science, informatics, work practice studies, media studies, product design, poetry, set design, music and performing arts. What they all have in common is an intention to contribute to the field of interaction design, to explore how people and digital technology can coexist and interact in better ways.

Sofia Dahlgren presents a close reading of a seminal work in new media studies, *The Language of New Media* by Lev Manovich. She demonstrates by examples from contemporary art how the narrativity/database dichotomy can be questioned and indicates what that means for the concept of navigation. Her main point, however, is that interaction design can be understood as the design of rhythm. This is a strong elaboration of what we might mean by temporal qualities of the digital material.

Peter Fröst has concentrated on issues of sketching and visual representation in design, drawing originally on *On Line and On Paper* by Kathryn Henderson but building his case for viewing real-time 3D visualizations as design artefacts mainly on a series of experiments in which such visualizations are used in collaborative design settings. Based on this experience, he identifies a number of desirable properties of the visualizations, including confidence, openness and sketchiness.

Ylva Gislén addresses the issue of narrativity in interaction design, following Umberto Eco's *Six Walks in the Fictional Woods* into literary notions such as fictional agreement, the reader's position, and spatial relations and discussing their implications for digital media. She warns against naive essentialist interpretations of narrative structure and advocates rich narrative spaces, »where the narrative elements and the courses of events can be negotiated, where conflicts can be formulated and dissolved, meanings multiply and overlap.«

Mikael Jakobsson discusses the design of virtual worlds. Drawing on the philosopher Michael Heim as well as on his own considerable experience of inhabiting and designing virtual worlds, he outlines a design position that transcends the dimension of realism versus fantasy. Rather than starting with the quasi-physical structures or the user functions of the virtual world, the design process takes its

point of departure on the level of social interaction. And, as Mikael reminds us, social interaction in virtual worlds is not a matter of simply transferring interactional characteristics from the physical world. To know that first-hand takes a considerable amount of lived experience, which is why Mikael's efforts at articulating his experience are so valuable.

Martin Johansson addresses the relation between ethnography and design in collaborative design processes, or, to put it more generally, how to make the essential design leap between how it is and how it ought to be. He introduces a design technique where ethnographic field material can be used as building blocks in collaborative design sessions, thereby grounding the design process in practice and facilitating active participation by analysts as well as by visionaries.

Per Linde concentrates on Anthony Dunne and the distancing qualities of interaction design, in analogy with the role of poetry in relation to prose. His concern is with the aesthetic qualities of the new spaces shaped not only by physical but also virtual materials. The design programme he proposes centers on creating mediators of experience rather than tools for instrumentality. The faith in the »users« and their ability to interpret rather than understand is high and hope-inspiring.

It will be apparent from reading the following chapters that the authors speak in very different voices, drawing on disparate traditions of thought. It will also be apparent that these different voices have one thing in common: they all search for their own partial answers to the question of how people and digital technology can coexist and interact in better ways. They are all searching voices within the emerging field of interaction design.

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INTERACTION DESIGNERS AS DESIGNERS OF RHYTHM:
A COMPARISON BETWEEN INTERACTION DESIGN AND NEW MEDIA

Sofia Dahlgren

INTRODUCTION

Besides that new media, just like interaction design, still is in its cradle the two have much else in common, dealing with similar if not the same kind of design issues. The author Lev Manovich writes in his book *The Language of New Media* that new media is »based on a number of various conventions used by designers of new media objects to organize data and structure the user's experience« (p. 7). Not only is the quoted sentence a concise and vigorous description of what new media designers do, but I also find the quote to be a quite essential description of what interaction designers do.

By focusing on three of the book's central themes—cultural interfaces, the info-aesthetics of narrativity versus database, and navigating through new media space—the aim of this text is to discuss some of Manovich's thoughts on new media and compare them to my understanding of interaction design. But first of all there will be a short introduction to Manovich's understanding of some new media conventions and terminology, which will be complemented by other new media authors' reflections. At the end of the text I will conclude the discussion by summarizing on the subject.

What is new media? The terminology that positions Manovich in relation to other authors

New Media consists of not only one, but many different cultural forms, which makes the subject rather complex when trying to get a notion of its essence. Various media like computer games, web sites, the Internet, computer multimedia and digital video are some of the new media forms that we are familiar with. While the authors of the book *Remediation* (Bolter and Grusin, 1999) claim that there really isn't anything new about new media, except the ways in which they refashion older media, Lev Manovich points out five different principles that he considers are new and significant. The principles are 1) numerical representation, 2) modularity, 3) automation, 4) variability, and 5) transcoding. By defining new media by what it can do technologically he falls into a trap which the authors of the book *New Media: A Critical Introduction* would call »technological essentialism« (Lister et al 2003). »There is a difference between looking, on the one hand, at why and how usefully certain concepts and ideas have arisen in efforts to define and understand new media, and assuming that such concepts are real features or even satisfactory description of the media in question, on the other hand. [...] Recognizing that a single media technology can be put to a multiplicity of uses, some becoming dominant and others marginal for reasons that have little or nothing to do with the nature of the technology itself, is one important way of understanding what a medium is.« (Lister M. et al. p. 13 -14).

In other words, Manovich's rather technology-focused principles are not meaningful to everyone in the same kind of way. For instance, numerical representation is rather marginal to a common user of a program, while it means everything to a programmer. This may be the reason that Manovich also suggests another more general definition of new media by describing it as consisting of two different layers, one »cultural layer« and one »computer layer.« Categories that he places in the cultural layer are for instance story and plot, composition and point of view, mimesis and catharsis, comedy and tragedy, while categories that belong to the computer layer are themes like sorting and matching, function and variable, computer language and data structure. This division between technology and culture is similar to how Lister et al declare their understanding of new media. Although the field is far from being fully identified they claim we face »a complex set of interactions between new technological possibilities and established media forms.« This idea of looking at established media forms when analyzing new media seems crucial also to Manovich (see Cultural Interfaces, below).

Compared to the book *New Media in Late 20th Century Art*, which is written from an art history perspective and includes a great many examples from performance, video art, video installation art and digital art, Manovich has a clearly technical approach. The way that the technical approach permeates the book is not the least in the choice of words. For instance, he uses the word »object« instead of words like »product«, »interactive media« or »artwork«, as if he wishes to address certain general principles in new media and thus look beyond its context of use or purpose. One may think that the word most naturally refers to something static and not at all to the temporal qualities that we experience in new media. However, knowing how the Russian avant-garde of the 1920's worked may help in understanding »object« as something temporal. Based on ideology, their purpose was to bring in ideals of rational organization of labor and engineering efficiency of factories and industrial mass production into their own work. The Russian avant garde artists used the word »object« to refer to their »production art« pieces, and considered art as an ongoing process. Manovich's reason for using the word »object« is to reactivate the systematic and laboratory-like experimentation performed by the avant-garde of the 20's, but also to connote to object-oriented programming. By doing so he once again underlines his computer science approach to new media. As this is a text about new media and interaction design, I will also address Manovich's thoughts on interaction and interactivity briefly.

While the authors of *New Media in Late 20th Century Art* describes interactivity as a »new form of visual experience, a new form of experiencing art that goes beyond the visual to the tactile« (Rush, M et al., 1999, p. 216), Manovich writes that he avoids using the word »interactivity.« All objects that are represented in a computer are interactive automatically, Manovich claims. Using the term is to Manovich meaningless as it simply expresses a rudimentary fact about computers. Other commentators (Jensen, 1999; Shultz, 2000; Huhtamo, 2000; Aarseth, 1997) are of the same opinion addressing that the term needs to be further defined if it is to be of any analytical relevance. Although I agree upon most of this, I am not quite comfortable with Manovich's description of all objects that are represented in a computer as being »interactive.« Watching a DVD movie on the computer does not necessarily mean experiencing more interactivity than watching a movie on TV. What about »digital«, could that be a more accurate description of this kind of representation? Lister et al argue that the term »digital« postulates a break between analogue and digital media, whereas no such thing exists. Some digital media are just reworked copies of old analogue media. The way that Lister et al have solved this maze of new but tricky words is not by totally avoiding to use the term »interactivity«, but instead dividing it into two associate clusters of meaning (one being in-

strumental and the second ideological) and moreover explaining the terms cultural and social history. Manovich does discuss the cultural history of human computer interaction (see below), but all too sparingly does he write about its ideological meaning beyond Russian film nostalgia.

CULTURAL INTERFACES

> The language of cultural interfaces is a hybrid. It is a strange, often awkward mix between the conventions of traditional cultural forms and the conventions of HCI—between an immersive environment and a set of controls, between standardization and originality. (Manovich, 2001, p. 91)

Cultural interfaces can have their origins in both HCI and other cultural forms like the printed word and cinema. Bringing in aspects of prior cultural forms simplifies the process of understanding a rather young medium better. Even though Manovich discusses cultural interfaces in relation to both HCI and the printed word, it is the screen that he pays the most attention to, including the film screen, computer screen, the television, the radar and the renaissance painting. He organizes them according to a genealogy, so that one screen type becomes the subtype of another. The interactive computer screen becomes a subtype of the real-time type (radar and real-time television), that becomes a subtype of the dynamic type (the film screen), that finally becomes a subtype of the classical type (the renaissance painting). No matter what type, the screen presents an image space that makes the subject more focused on that space than of the physical space that surrounds him or her. Manovich claims that these kinds of so called »dioptric arts«, in which the subject experiences a split from being present in a physical space and a screen image space at the same time, imprisons the subject (p. 103). The viewer of an image needs to be immobile in order to get a proper experience of it. With VR it is different, he claims, but no less imprisoning. Visually, the screen disappears together with the physical world, while the image space dominates in total. The immobility disappears, as it requires the user to move around physically in order to move in the virtual world. At the same time, VR imprisons the body even further as the body becomes nothing but a tool to run a program with, like a giant mouse. Similarly, Anthony Howell discusses the negative effects of bodily interaction in performance art (Howell, 1999).

> […] the participatory element depresses cognition in the spectator. Its invitation beguiles him into the enactment of obedience. Instead of actively thinking about what is going on (i.e. creating a subtext to the action) the spectator passively performs the action, such as clicking on a mouse or accepting a sweet, simply because it is demanded by the apparatus or by the performer. The danger is that while a certain liveliness has been conferred to the artwork, the spectator has been reduced to an automaton. (Howell, 1999, p. 63)

While the performance artist Stelarc, famous for creating cybernetic art (robots, cyborgs etc.) reflects on the positive aspects of using technology as it enables us to go beyond the borders of our bodies (Lenas, 2002). Walter Benjamin and Paul Virilio have yet other thoughts on why physical interaction shouldn't be encouraged, at least not when acting from a distance. According to them aggression is potentially present in touching when acting on distance, as the distance creates a gap between the subject and the Other. It enables the subject to treat the Other as object, it enables an objectification. This is a standpoint that Manovich agrees upon, claiming that touch opposition is something we should retain (Manovich, 2001, p. 175).

Benjamin and Virilio's deduction that touching on distance connected to technology becomes a generator of aggression does not only seem irrelevant but also dogmatic. However, Bruno Latour adds some complexity into this discussion. Instead of pleading that the relation consists of the active human as subject and on the opposite side the passive technology as object, Latour argues that a mediation takes place in the meeting between different actants (technology or humans), which creates a third mediated actant. This last actant is the one that really matters. He gives an example in the discussion between the ones who are against arms and those who advocate arms. While the former claims that »guns kill people« and the latter argues that »people kill people«, Latour does not agree upon any of these statements, but concludes that »people with guns have the potential of killing people.« Thus, instead of getting totally dogmatic in this old (if not ancient) discussion about the consequences of technology as extensions of our physical bodies, we may want to consider that some responsibility lies on us as new media or interaction designers in trying to foresee and prevent unpleasant mediated actants.

Manovich chooses to discuss cultural interfaces mostly as visual interfaces, although haptic interfaces are mentioned. Visual interfaces are dominant within new media and probably one of the issues that really signifies it, even though it sometimes includes a tool, body or mouse to run it with. But perhaps this is the border to where new media becomes something else, like, say, interaction design…? Seeing it

from an interaction designer's point of view: to not include the activities connected to the visual may mean to exclude the wholeness of new media pieces (see Viola's *Going Forth by Day* below). Interaction designers may want to develop their own suggestions of cultural interfaces to complement the ones Manovich suggests for new media. Focusing not only on activities connected to traditional graphical user interfaces (GUI), interaction design may also want to include cultural interfaces and technologies that involve the whole body and space, like for instance dance, performance and theatre improvisation.

NARRATIVITY VERSUS DATABASE

As mentioned before, there is especially one dioptric art that Manovich believe could inspire in the design of new media, and that is film. Future new media will have a greater resemblance to film, he claims, and the narrative premises with inspiration from film can become more complex than they are today. Manovich refers to one specific movie that he considers more inspiring for new media than any other and that is Dziga Vertov's *Man with a Movie Camera* from the 1920's. The movie is an avant-garde documentary about a man standing behind his camera, filming all that he sees, but as viewers we also see the audience, what the man sees through his camera lens and even beyond the man's perceptive reality. Jumping between different narrative levels, layers and perspectives the film expresses Vertov's idea of trying to capture all visual effects that our own human vision can not see. This rhythmically moving camera technique that Vertov developed is called a kino-eye technique, and it is this kind of technique that Manovich suggests could inspire new media designers.

Vertov's movie may prove that visual effects can be used to create a meaningful artistic language. However, it would have been interesting to read further about Manovich's visions of how Vertov's art piece could be navigated through if it were to become more interactive, or perhaps how it could be complemented by tools supporting the visual transforming it to something beyond merely visual. Even though he sees possibilities in how aesthetics can be created through the navigating between perspectives, layers and narrative levels, he does not give any concrete examples of what for instance would motivate a user to interact so that it creates what he calls a viewer's or user's experience. The movie is just an example of one artist's language among many others, which may be inspiring in the design of certain kinds of new media. How it inspires the production of interactive new media in general I have a hard time imagining. But if I were to speculate on the subject, it seems as if Manovich has chosen to describe one kind of interactive form, which resembles

activities similar to the use of a control panel rather than navigating through the simulations of a 3D world. Navigating through the Vertovian space would in that case be similar to using an editing board, which if it were to be considered fun or interesting would include a task that would motivate the user to explore. The kind of task could perhaps be comparable to the one that Tom Cruise's character gets in the movie *Minority Report*. By navigating through different narrative layers and by using many different screens, he searches for valuable information under time pressure. These kinds of activities are what most users (or players) of interactive narratives (or games) enjoy. New media that include many screens are not specifically immersive in a physical sense, but may be in a psychological sense.

These different kinds of immersive experiences in new media are identified by Manovich as two different new media forms: 1) simulation versus representation and 2) representation versus control. In the first dichotomy representation is defined as various screen technologies: »a rectangular surface that frames a virtual world and that exists within the physical world of a viewer without completely blocking her visual field« (Manovich, p. 16). The opposite word simulation »refers to technologies that aim to immerse the viewer completely within a virtual universe« (p. 16). In the second dichotomy, representation versus control, representation stands for the image as an illusionary fictional universe, whereas control stands for a simulated control panel. When experiencing narrative stories in movie theatres we are familiar with how simulation (1st dichotomy) and representation (2nd dichotomy) immerse the viewer both by taking up physical space and on a psychological level by bringing the viewer into an illusionary fictional universe. While in so-called database activities we recognize representation (1st dichotomy) by the non-immersive screen and identify the similarities to a control panel (2nd dichotomy) in the interface. These two different forms called database and narratives are identified as two opposites. This brings us into another of his theses—how a database could equal narratives and become a part of an artistic language.

> We need something that can be called »info-aesthetics«—a theoretical analysis of the aesthetics of information access as well as the creation of new media objects that »aestheticize« information processing. (p. 217)

This info-aesthetics that Manovich describes is discussed and analyzed through the cultural lenses of cinematic narrativity and of database.

Narrativity and database are according to Manovich not only two opposite poles, but even natural enemies. The traditional definition of database according to computer science is »a structured collection of data« (p.218). The data can be

organized differently (i.e. hierarchically, as a network, relational, and object-oriented), but should be organized for fast search. Some of this data is directly accessible to the viewer as part of the experience. According to Manovich the database ought to be considered as a cultural form in its own right, just like any narrative or architectural plan is considered to be a model of what the world is like. A database, representing a list of items, has no logical order in terms of development as its opposite, the narrative has, creating a trajectory of cause-and-effect (p. 225). To design database can for instance mean to construct »the right interface to a multimedia database«, while designing narrative could mean to design »navigation methods through spatialized representations« (p. 215). In Manovich's reasoning, the narrative equals the syntagm, which is defined as a »combination of signs, which has space as support« (Barthes, 1966, p. 58), and the database equals the paradigm defined as »the units which have something in common are associated in theory and thus form groups within which various relationships can be found« (Saussure, quoted in ibid, p. 58). In natural language the syntagmatic dimension is the linear sequence that strings elements in an utterance together, while the paradigmatic dimension is in »each new element (that) is chosen from a set of other related elements« (p. 230). Examples of paradigmatic groups in natural language could be nouns, verbs or synonyms. No matter what the groups are they remain virtual and exist only in the reader or writer's mind, while the sentence printed on a piece of paper or spoken is material and real. In new media, Manovich claims, the relation is the reversed: »database (the paradigm) is given material existence, while narrative (the syntagm) is dematerialised«(p. 231). Database is real and explicit, while narrative is virtual and implicit, Manovich claims. The narrative of a film is considered to be only one trajectory out of many possible narratives through a database, while the interactive narrative of, e.g., a computer game consists of many possible trajectories through a database.

Using the words »natural enemies« and »opposite poles« are to me a bit strong, as it may indicate that they cannot co-exist within the same timeline and complement one another. I believe they can, which I will argue for by discussing new media artist Bill Viola's work *Going Forth by Day* (Viola, 2002). Viola's digital-image installation explores human existence on themes like individuality, society, death and rebirth. In the rectangular room where the installation takes place five different scenes are projected simultaneously, covering all four walls in the room. Seeing it for the first time it is impossible to pay attention to all details going on at all the different projections, therefore visitors seem to stay in order to experience the piece a second time or even a third time to experience it differently. Zapping between the different projections by looking in different directions, experiencing the partial as

well as the whole is part of the experience in total. Making the choices of how to perceive the piece through physical interaction I believe could equal the potential choices existing within a paradigm, while watching one single projection at a time, on the other hand, would be to experience it syntagmaticly. So, enemies or not, they do not necessarily exclude one another but can co-exist and add value to the experience of the whole, even though human perception has limitations that prevent us from being able to focus on all narrative trajectories at one time.

Having discussed narratives and database in relation to more traditional narrative forms, what does this possibility of choosing between different narrative trajectories mean to the player of a game or user of a web page? When it comes to the use of a traditional web page where all essential information needs to be accessible I agree that the database is explicit. When it comes to games the situation of being able to choose between many different possible alternatives is the same although perhaps not quite as obvious. There is an »algorithm« that needs to be uncovered in games. Thus, the paradigm (or database) needs to be well hidden for the player intentionally, as part of the game's playability, which is why some of the database choices are not even created as explicit choices. In other words, players probably don't think of database being explicit and material quite as simply as Manovich claims, as hiding at least some of the choices is often part of making the game interesting. There is another important ingredient in the database versus narrativity discussion, and that is to navigate and explore.

NAVIGATING THROUGH SPACE

Looking back in history, to organize information or data spatially is nothing new. Although practiced much earlier, it was in the 13th century that advanced mnemonic techniques using architectural visualization were developed by Thomas of Aquino (Manguel 1999, p. 67). This was a method that was successfully used for hundreds of years.

New media spaces can either be representations of physical worlds or abstract spaces of information. No matter what, »where there is space there is navigation« (Manovich, 2001, p. 252). Whereas interaction design does not necessarily deal with representational forms, new media is by definition representational. However, navigation is something the two have in common. Along with database Manovich suggests that we consider navigable space as a cultural form in its own right. Several examples from other cultural forms of how human relates to space are given. One example of spatial attention defined in the beginning of last century by the art historian Alois Riegl is to perceive space haptically or optically (Manovich, 2001,

p. 253), while another example given by the installation artist Ilya Kabakov is to while wandering focus on the whole or actively and well-aimed take in the partial (Manovich, 2001, p. 267).

When it comes to navigating through new media spaces the journey has often been described as seeing through the eyes of the Data Dandy. Originally introduced by Geert Lovink, the Data Dandy has clear reference to Baudelaire's flâneur. Walking through crowds in Paris, the flâneur records and erases the faces and bodies of the women he passes by, having an imagined affair with each and every one of them. He remains anonymous and creates his own subjective space as he unceasingly moves through space. Similarly the Data Dandy considers the net to be what the Avenues of Paris were to the flâneur. When navigating on the net »the affairs«, or what Kabakov called the partial, need to be made interesting as independent objects, while the stroll in between »the affairs« is less interesting or even pointless, as navigating to the new page ought to take no time at all. Navigating in a fictive virtual world, on the other hand, is different from navigating on the net in that the stroll from one place to another needs to be made as interesting as anything else is in the fictive world.

A parallel to the relation between navigating in a fictive world contra navigating on the net may be drawn to Barthes' relational description of classic poetry versus modern poetry (Barthes, 1966). I believe it may even describe the relation between narration and database more clearly. Barthes states that the words of classical poetry (and prose or narration) create a connecting link more than they signify things. These words have lost their saturated meaning and the partial serves a less important role—almost reduced to means of transport—to the benefit of the context in its whole. In modern poetry which in this case is compared to surfing on the net (or database navigation), the Word plays a central role. The links are not cut off, but the connecting-points are isolated. The word stands alone and is encyclopedic »as it contains all interpretations that a relational style would have forced it to choose between« (p. 36). Modern poetry is however different from net or database navigation in at least two matters. First, modern poetry does not include any explicit choices like the ones database produces, secondly, the relational condition between the words does not really »fascinate and is consumed by the reader with curiosity« (p. 36) as one may assume it does when reading (high quality) modern poetry. Too rarely do we experience the relation between links fascinating in database navigation, but perhaps in art it is likely that we experience it more often. Once again I would like to mention Viola's installation as an achievement in that direction.

When navigating through fictive spaces of, e. g., computer games, instead of using the terms »narration« and »description« as in narratology the terms »nar-

rative actions« and »exploration« are suggested to substitute the former. As with traditional poetry or narratives, the passages from one dramatic space to next is just as important as the narrative events themselves. Navigation is described as a way of moving the narrative forward (p. 247). Manovich writes: »If the player does nothing, the narrative stops. From this perspective, movement through the game world is one of the main narrative actions.« Since more and more games seem to advance by including autonomous features, Manovich words are not quite true. In the game *The Sims*, for instance, the autonomous characters will create narrative events even though the player is passive. »[…] in many American novels and short stories, narrative is driven by the character's movements in the outside space. In contrast, nineteenth-century European novels do not feature much movement in physical space because the action takes place in a psychological space« (p. 271). According to Manovich this latter kind has not yet been explored thoroughly in the design of computer games.

UNDERSTANDING THE RHYTHM OF INTERACTION DESIGN

There is nothing odd about letting technology have a strong influence in the naming of genres. For instance in electronic avant garde music of today, there are genres like »electronica«, »electro-clash«, »clicks and cuts« and »glitch« that all origin from technology. Quite similarly, Manovich proposes that we rename new media to »software studies«, which has clear connotations to information technology. As media studies of today does not include the study of the programmable, while computer science does not include studies of other media, he suggests that software studies be a science of its own. In the book *Bringing design to software,* Terry Winograd proposes that the word »software« associates to conventions such as application functionality, structure of program components, and the look and feel of an interface (Winograd, 1996). In other words, if adopting Manovich's suggestion there may be a risk that one associates »software studies« to traditional GUI conventions. But perhaps »software« could also allude to the »soft« values of the Enlightenment Project, such as art, aesthetic ideals, ethics and politics, in prior to the existing »hard« values based on natural science and economic growth. However, neither software nor its opposite hardware does necessarily connote to human perception. A word that explicitly is associated with perception is »reality«, thus the already existing dichotomies—virtual and augmented reality—is not only less technical, but probably a more interesting way of discussing the matter. Even though virtual reality (and new media) deals with a reality that is situated

»there« in opposition to »here«, considering the design of activities taking place in the physical space is just as important, no matter how tactile and subliminal these activities are. Bill Viola's new media piece *Going Forth by Day* is just one example of this. Interaction design concerns the design of both these realities, which is why an interaction designer may find it pointless to speak of new media without considering human perception.

Manovich writes about the interface as the major form of new media, but it seems as if he ignores the interactive aspects that may be connected to new media, separating the database from the narrative, ignoring that interacting with an interface creates a mediated form that goes beyond its interface. This rhythmical and dynamic understanding of interaction that include an evolution over time and in interaction with a user, that in interaction design is called dynamic Gestalt (Löwgren and Stolterman, 1998) is something that Manovich does not really discuss. However, he does quite poetically describe how a DJ works, which is the closest he gets to describing the presence of dynamic Gestalt in new media design contexts.

> The DJ best demonstrates its new logic: selection and combination of preexistent elements. The DJ also demonstrates the true potential of this logic to create new artistic forms. Finally, the example of the DJ also makes it clear that selection is not an end in and of itself. The essence of the DJ's art is the ability to mix and select elements in rich and sophisticated ways. In contrast to the »cut and paste« metaphor of modern GUI that suggests that selected elements can be simply, almost mechanically combined, the practice of live electronic music demonstrates that true art lies in the »mix«. [...] The DJ's art is measured by his ability to go from one track to another seamlessly. A great DJ is thus a compositor and anti-montage artist par excellence. He is able to create a perfect temporal transition from very different musical layers; and he can do this in real time, in front of a dancing crowd. (Manovich, 2001 p.135, 144)

The metaphor may be idealized in many ways, but it is yet quite illustrative when discussing the art of interaction design. And hopefully interaction design will, differently from traditional HCI, position itself as carrier of artistic values similar to the ones discussed within new media art. Values that for instance may be associated to poetry, metaphysics and aesthetics. But yet different from new media, allowing other interfaces than the screen-based ones to come through.

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DESIGN OF REAL-TIME 3D VISUALISATIONS AS DESIGN ARTEFACTS

Peter Fröst

ABSTRACT

The importance of the use of design artefacts in collaborative design involving users and other stakeholders has been pointed out by a number of authors. In these design settings the role of design artefacts is to support collaborative work, dialogue and innovative thinking. This article focuses on how to visually design real-time 3D representations in order to be productive design artefacts in collaborative architectural design environments. The way in which real-time 3D conveys information and generates reading is discussed and a number of categories for mapping real-time 3D representations are proposed. Finally, based on the experiences from a row of presented projects, a list of mementos for future work is formulated. Real-time 3D representations used in early phases of collaborative architectural design should take the following into account. *Detailing:* have an information density coordinated with the design phase. *Simplification:* eliminate features of real world complexity—support creativity. *Appreciation:* support a positive attitude among participants. *Confidence:* representations must be correct and trustworthy. *Openness:* have a conscious use of analogies and metaphors. *Leaving traces:* you must able to influence and »mark« the representation. *Sketchiness:* be visually open—support multiple interpretations.

DESIGN FOR COLLABORATIVE DESIGN

In our research group at the Space and Virtuality Studio in Malmö, we have explored the possibilities to use collaborative design to identify customer needs as input to the architectural programming and design process in workplace design (Johansson et al, 2002). This suggests a design process different from the standard praxis today. The concept of »customer engaged design« is based on a process that is workshop-driven. It includes active collaboration between customers, users, different stakeholders and designers and develops new design concepts through joint interaction. By collaborative observation, inquiry, design and evaluation, the participants play an active role in exploring existing workplaces and in making new work environments.

Real-time 3D Representations as Design Artefacts

In the workshops participants use various design artefacts, such as video, design games, scenarios and interactive real-time 3D visualisations. The design artefacts play an important role by supporting dialogue, collaborative work and innovative thinking in the design process. They are designed with the purpose of facilitating common understanding of the design problem and joint design work. Opening up the architectural design process by inviting participants who are unfamiliar with design work, »complicates« the design. Brandt emphasises the role of representations in keeping continuity between events in a workshop-based design process (Brandt, 2001). Understanding each other is often difficult when the participants have various competencies and perhaps various professional languages. When a diverse group of stakeholders is involved, all with their own expertise and various interests, it is a challenge to create a common language, a ground from which a design dialogue can spring. Ehn (1992) suggests seeing multi-person design sessions as a meeting of language games, and argues for the need to create shared design artefacts that can span the gap between these language games. Design artefacts are with this understanding primarily means to support users and designers in their dialogue in a design situation and visualise possible futures. Design artefacts should consequently not merely be utilised as illustrations, or creators of »true« images in the design process.

Focus and Structure of this Article

Looking at the design process with a metaphor from theatre you can say that we are aiming at supporting the actors with components and props for creating the staging of a play that they know beforehand. In this endeavour, it is very important how these props are designed in order to give the play an appropriate setting. This article focuses on the question how to visually design real-time 3D representations in their role as design artefacts. In a series of collaborative design workshops, we have introduced a computer-based tool named ForeSite Designer. The idea and main feature of the application is to make a fast connection between 2D layouts of architectural space that you can interact with and a real-time 3D representation of the same space. The computer application has proved to be a well functioning design artefact in collaborative design work (Fröst 2002). This article will address the role of real-time 3D visualisations produced by ForeSite Designer, and how they should be designed.

First, I will briefly outline some roles of visualisations in today's architectural design process. Secondly, I will tell the story about how we have tested using real-time 3D visualisations as design artefacts and describe experiences from a number of performed projects. Thirdly, I will discuss how architectural visualisations in general and real-time 3D in particular communicate. I will also try to identify some suitable categories for understanding architectural real-time 3D representations. Finally, a list of mementos for future work is formulated.

Definitions

With 3D visualisation I mean the quality of having three dimensions, which for example distinguish a drawing, which is in 2D, from a perspective in 3D. Real-time 3D means that you can move around freely in a 3D representation, it is an interactive screen-based animation where the observer has the ability to control the virtual point of view (which distinguishes it from a film or fly-through where the point of view is predefined and controlled by someone else). Virtual Reality (VR) is a technique in which you can move around and interact with a 3D representation. It is often also named a »computer-generated immersive environment.« You can experience VR on a screen but also with display devices such as a HMD (Head Mounted Display). A «Cave«, or grotto, is a very technically advanced and expensive system, where images in scale 1:1 are projected within the walls of a box that surrounds you almost completely. By letting the computer generate two images, one for each eye, and by wearing special glasses, the images in a Cave are experienced in stereovision (Cruz-Niera et al., 1993).

SOME ROLES PLAYED BY VISUALISATIONS IN ARCHITECTURAL DESIGN

You can assume that the introduction of new interaction technology in the building sector will in the long run completely change the way planning, designing, bidding, construction and maintenance will be performed. Customers and users will raise new demands for influence and information inspired by the possibilities of new technology and media. The design process will accordingly be re-designed; collaboration and visualisations will be a part of every phase of the process.

Architectural Visualisations

Architects constantly work visually with sketches, drawings and different forms of visualisations. Simple »work« visualisations and mock-ups are often made with the purpose of understanding one's own design better and for internal communication in the professional design team. More elaborated visualisations, renderings and perspectives are produced in order to communicate in some way with the external world. Even these visualisations are produced with different levels of detailing. When the main purpose is to convince clients, investors, customers, administrators etc, you often make very »selling« visualisations conventionally grounded in a naturalistic manner, where everything is aimed at assuring the viewer of the qualities of the project. But when you want to present a vision of how something possibly could be, you are most likely to choose a more open and sketch-like approach. In that way you can persuade without having to tie yourself to a specific definite solution.

During the last decades architects have made digital 3D visualisation an integrated part of their visualisation practice. From a very coarse, sterile and artificial look (which had its effect in the »high-tech« associations), the computer-made visualisations have developed a new creative visual language and expanded the fields of architectural renderings. Many real-time 3D visualisations can nevertheless still be regarded as being in an immature stage, heavily leaning on naturalism, probably due to the fact that this still is an rather unusual way of visualising architectural projects.

New Users—New Visualisation Tools

The design process in itself reflects the types of tools and media that are used. The 2D drawings that are created to describe the building by the architect make very

suitable tools for the professional actors (Campell and Wells, 1994). An expanded interest group, including clients and the future users, also means new demands on the visualisation tools. These must provide the possibility to create an understanding for users and simultaneously work as an effective design tools for those responsible for the design work. Different forms of visualisations, such as 3D perspective renderings and scale-models, have therefore been developed to create an understanding and communication around the planned construction work. To further facilitate the communication and the information handling, new digital tools have been introduced, such as real-time 3D visualisations and Virtual Reality. These facilitate an increased understanding for those people who are not used to the traditional tools of the construction process.

The external participants' need for visualisations is most crucial in the early phases of the design process. This is particularly true in a collaborative design process where customer and users themselves are engaged in the design work with sketching of different kinds. An inequality in the possibilities to grasp and understand the design task creates inequalities concerning the influence on the design process. Here it is important to develop easy methods to transform sketches into visualisations for presentation usage, without having to spend too much work on producing an arranged, by necessity restricted, selection of the presentation material (Wikforss, 1978).

A JOURNEY FROM CAVE TO COMPUTER GAME BASED REAL-TIME 3D

This is part of a story about how a cross-disciplinary research team has worked together for several years on the development of new design artefacts for collaborative architectural design, focused on how to involve users in the design process. As we work within an action research format, the journey has gone from workshop to workshop were we have developed and tested real-time 3D visualisations as design artefacts. In the following I will describe the main events in the development of the artefacts and experiences from a number of performed projects.

Chemical Department—Collaborative Design in a Cave

We started in the spring of 1999 with a project where we examined the possibilities to use Virtual Reality in a collaborative design process involving end users and



other stakeholders. It was carried through in co-operation with the Chemical Department at Lund Institute of Technology/Lund University (Fröst and Warrén, 2000). Design proposals from two preliminary workshops where modelled in 3D, simulated with VR software, and at two different design events brought into a VR Cave at Chalmers University of Technology. Two different visual levels were tested. At the first visit the interior walls, glass walls, doors and ceilings were represented as single coloured surfaces. Neither details such as textures, joinery or fences nor installation equipment, electric fittings, plants or people were represented. There was directed light but no shadows or reflections. The building was situated in an empty space without surroundings. The representation could therefore be characterized as a sketch representation made of cardboard but represented in digital form. On the second visit two weeks later there were some changes added to the representation. Characteristic surfaces were mapped with textures and pictures as carpets, details of cupboards, blackboards and so on. Through the windows you could see the actual view outside the building.

The participants did not pay much attention to the two different levels of detail presented in the Cave visits. They were very focused on their own design task, they wanted to know exactly how their laboratories were going to be, and were less observant on the VR tools as such. No bias discussion about irrelevant information in the representations emerged. The participants felt that as not everything is finished, it was possible to change the representation. A conclusion that we made was that in this case, the simple and low detail performance of the sketch-like VR representations actually promoted dialogue and creativity.

Based on our experiences from the Chemical Department project, we concluded that the VR system in the Cave was too complicated and difficult to access for the workshop format. During the last years of the 20th century, large investments where made on building Cave systems. The tendency today, however, seems to be that Caves are considered to be too complicated and expensive to be used to their full potential within the field of architectural visualisation. The Cave is optimised for individuals or small groups. We wanted something that could work with several small groups simultaneously and for larger groups at presentations, etc.

Hardhat Designer

This turned our interest to real-time 3D engines based on computer games. At a visit to Martin Center, Cambridge University (UK), in the summer of 2000, we were introduced to a representation of a complete building (10,000 m^2), built on the computer game platform Quake (Richens and Trinder, 1999). We found that several other modern computer games also provide advanced real-time 3D visualisations and rich graphical environments. They are widespread, cheap and easily accessible and can be used on a »standard« PC. Advanced modelling tools are available for free. The game technology develops and improves very rapidly through commercial and non-commercial processes (there is a whole Internet community working hard, sharing information). It is easy to learn and fun to use, hence stimulating creativity as well as fast learning. We selected the game Half-Life because it is technically open to work with. It also supports different kinds of modelling tools such as Worldcraft and Quark. The method of building libraries with prefabs makes it easy to customize for every session and objects can easily be reshaped and exchanged.

Our interpretation of the Chemical Department project also pointed to another feature that we wanted to achieve. Inside the Cave the environment immerses you. You get absorbed by what you see and talk about everything that shows up. Outside the Cave, around a common worktable, the discussion turns to a more traditional work mode. You analyse, sketch, measure, and solve problems. The interaction be-

tween these two different situations constituted the design-event. This indicated to us that VR immersiveness provides a good common ground for evaluation of design proposals. However, we found it less supportive for developing new design suggestions. To be able to be active in reflective design discussions, we wanted to support the participant's needs to share a birds-eye-like observer perspective. This enables them to grasp a conceptual totality, which is not available when immersed in a particular design vision. This observation resonates with the shift in user perspectives often employed in computer games (Walther, 2002). When one is engaged in combat, the game offers a first-person point of view, but when you are laying out the strategy for how to go on and therefore engaged in rational contemplation, the game shifts to plan or isometric mode.

In the autumn of 2000 we were tutors in a student project at Malmö University where one of the students in Interaction Technology developed a design prototype, based on the computer game Half-Life, into an extremely »easy to use« digital modelling tool called Hardhat Designer (Fröst et al., 2001). Hardhat Designer enabled the users to »sketch« architectural layouts on a playground in 2D on the computer screen. That means that you have a repertoire of building elements, simplified projections of the Half-Life components, to place, move and rotate on the playground. This 2D layout could then instantly be exported to a real time 3D world in the computer game Half-Life. Like in several other modern computer games, we found the visual qualities in Half-Life sufficient for architectural representations, at least when the design is in an early schematic phase. Half-Life produces a lighted real time 3D environment with low-detail geometry and high quality texturing. We had now succeeded in making a rapid connection between 2D layouts and a real-time 3D level in Half-Life. In Hardhat the 2D interface gives a good overview for collaborative design decisions and the real time 3D a good environment for evaluation. In the first workshops to be carried out, the 3D world that we used was still very coarse, with elements taken directly out of the Half-Life component library. But the idea of a very rapid visualisation tool turned out to be successful. The possibility to view your design proposals instantly in real-time 3D proved very powerful.

Telia Research—Hardhat Premiere

The first design event intended to test Half-Life as a visual representation tool was carried out in January 2001. It was performed together with Telia Research (a research organisation within the Telia company) and a group of IT companies in a scenario workshop. We organised a visionary workshop with the task of building and visualising technical and spatial solutions, which could support collaborative



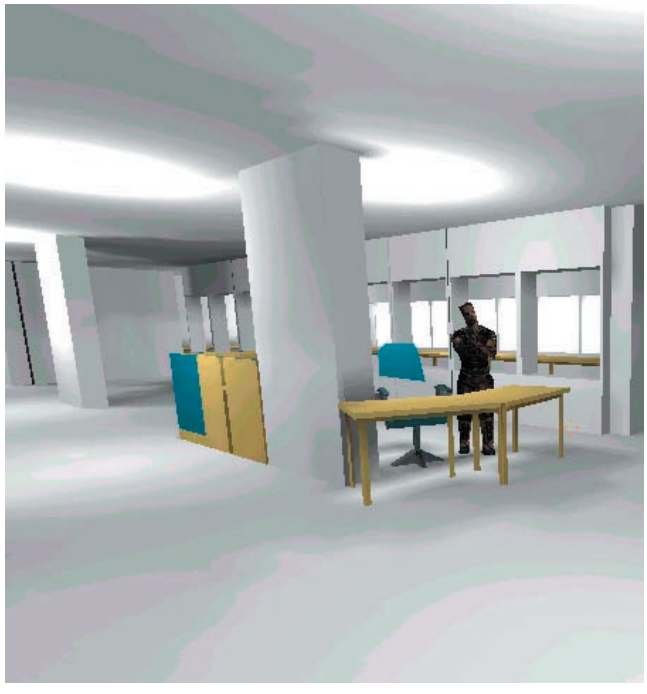
office work in the future. Thanks to Hardhat Designer, it was possible to immediately view what they had built in real-time 3D. Subsequently, a presentation was carried out on a large screen display where you could act before the real-time 3D world in scale 1:1, as if on a stage.

The detailing and visual quality of the real-time 3D worlds generated from Half-Life seemed to be reasonably adequate for the chosen tasks. Negative comments from users mainly concerned furniture design and lack of daylight. The furniture in this early version of the program was chosen directly out of Half-Life's standard, rather game-looking library. Half-Life seems to be designed primarily for interiors and a »garage« character was apparent in some of the built worlds. Despite this, an important quality in the Half-Life VR environment is the ability to very quickly produce artificially lighted spaces. Light has the power to make spatial representations »come alive« without having to overload them with irrelevant details. The participants regarded the real-time 3D spaces in our examples as comprehensible and having »character«, although the walls and furniture indeed were very simple.

Vasakronan—The Garage Revisited

A second workshop was performed in March 2001 with the real-estate company Vasakronan, together with one of its tenants (an R&D company housed in a traditional office building in Stockholm). After a series of introductory exercises the task here was to use Hardhat Designer to model a part of the participants' work environment. The chosen part had been identified earlier in the workshop as being in need of change. Within this part spatial reorganizations were outlined. The real-time 3D Half-Life worlds were then displayed on the wall and used at a presentation session in the end.

Based on experiences from the workshop with Telia, the visual design of the 3D world in Half-Life was changed and improved in some ways. The Half-Life furniture was exchanged for a set of more realistic modern furniture. The surface or »playground« on which you build in 2D could now easily be varied in size up to 80 by 120 meters. You could also display a grid and a ruler.



The participants worked together in groups of two. They had a hard-copy of the actual floor plan as their point of departure and started by making a 3D representation based on it. The exercise went on for about two hours. During this time the groups (unskilled in 3D modelling) managed to schematically build workspaces of about 3-400 m^2, furnish them and in some cases change and even add new rooms. To us this exercise showed that Hardhat Designer made it possible for unskilled users to build and manage rather complex 3D worlds in a relatively short time. Feedback from the users also forced us to start developing the visual appearance of the 3D worlds. The garage metaphor had reached its limits and we now started to look for more open manners.

ForeSite Designer

The interface of Hard-Hat Designer was still very primitive and difficult to work with and we therefore started a project where we developed a whole new interface, ForeSite Designer, FSD (Fröst, 2002). The idea was to make it possible to work with an unlimited repertoire of 2D images and to add several functions.

When using FSD you start by choosing a specific »playground.« The playground can be everything from an abstract empty scene to an elaborated architectural space. On this playground you are able to work with different prefabricated components. They can be acquired from the FSD standard component library or from a library that is built and tailored for a special occasion. Now you merely click on the playground where you want your component to be placed. It is possible to move, rotate and copy the component. You can also draw lines and squares on the floor to mark certain areas, measure distances between components, write down notations or messages, and so on. The playgrounds can have default lighting but it is also possible to actively insert light sources and in that way use light as a building element. Visually, the real-time 3D worlds generated by FSD were also developed compared

to Hardhat Designer. We now started a series of experiments with different designs of 3D environments and components.

Interactive Institute—Testing ForeSite Designer at Home

ForeSite Designer was first tested at an internal workshop in the Space and Virtuality Studio in June 2001. It was performed in order to examine how to arrange for a clearly understandable entrance to the studio that also worked as an interactive display area for the research that was carried out there. The participants worked in groups with the commission to identify qualities that were important to embrace. A real-time 3D representation of the studio had been prepared in advance and the commission was to develop the representation with devices that in some way symbolised walls, furniture and technology that could be integrated into the studio space. The representation was designed with pencil and watercolour textures. The walls and furniture were non-textured single-colour elements. The exercise took place inside the actual space where change was intended. This means that the participants were surrounded by real world while they had a 3D representation in their computers to work with.



An interesting observation was that the design discussion and work with the 3D representation tended to be more unbounded and imaginative than was the case when the real space was used as design artefact. The participants did not surrender to the limitations of the real space. A possible explanation is that the representation was low-detail and sketch-like and hence provided a distance from reality that was productive for the design dialogue. From this exercise we learned that it was very positive to apply textures in the representation, although they did not carry any information of the actual space but merely symbolised surfaces. We also found that windows with views onto the outside world were very valuable for orientation and reference to the internal space. The representation was not built with accurate

measures, which proved to be a disturbance when changing from real world to 3D representation.

Chalmers Studios—Developing Visual Features

Together with the Department of Architecture at Chalmers University of Technology, Göteborg, Sweden, a two-day workshop was conducted in September 2001 with the goal of finding new solutions to the studios for architectural education. The architectural students were offered ForeSite Designer as a design artefact. The visual design of the world prepared in Half-Life was not related to a specific building. The aim of the workshop was only to develop conceptual solutions focusing on functions, qualities and their relation to each other. Consequently, the real-time 3D environment in the Chalmers studios case was designed with the purpose of being non-realistic. A 25 by 25 meter playground with a modular system of one meter as floor pattern was prepared. There were no roof or exterior walls, the setting was



somewhere under the night sky. Components and furniture had an abstract »Mondrian-like« look with sharp volumes and primary colours.

The users in the Chalmers case were students of architecture, and as such experienced in discussing design matters and visual representations. The students are trained to come up with individual ideas on spatial solutions quickly, and the challenge was to make them unite in the groups. The building of the real-time 3D environment became a meeting place for their different ideas on how to design their own studios. It was apparent that they used the visualisations as generators of common spatial ideas. The purposely-artificial design of the real-time 3D was accepted as being non-realistic and as such appropriate for this conceptual exercise. We learned that it was possible to work with a very abstract visual appearance if it was properly designed. We also learned that it could be difficult to foresee how participants will use a design artefact such as ForeSite Designer. The representation itself generates new ideas. This should be anticipated and incorporated into the design of the representation.

The Experimental Office—Multi-Competence Design Facilitation

To have several different partners with different interests and competencies collaborating in a future-oriented design process puts certain demands on the design of the process and the design artefacts used. The Experimental Office project was conducted as a series of three workshops during the autumn of 2001. The commission for the participants during the third workshop was to build concepts for an office of the future. The preconditions were that a specific space of approx. 170 m^2 was given as the place where the Experiment Office should be built and realised. The premises where modelled as an actual 3D representation with doors, windows etc. Through the window you could look out onto the surrounding buildings and environment. Interior walls, furniture and other accessories were at hand, built as scale models with stylised colourful geometries. The technical support systems were visualised in two ways. Systems with physical representations were visualised as building components, such as a wall with an animated texture representing the large projection screen for remote collaboration. Systems without obvious physical representations, such as a wireless local area network (WLAN), were visualised as signs.

ForeSite Designer proved to be an appropriate design artefact for supporting the dialogue and expanding emerging ideas in this context. During the relatively short exercises, the participants succeeded in collaboratively building elaborated spatial



configurations, where space and technology were integrated in a meaningful way. One conclusion was that the commission to the participants to develop scenarios for their built spaces forced them to combine different ideas, to negotiate and prioritise. When the scenarios were acted out in the real-time 3D world the integration between space and technology was actually tested. The possibility to stand up in front of a real-time 3D representation and invite your fellow participants for a walk »inside« the planned workplace is a very powerful feature also in early stages of the architectural design process.

The adding of views of the surroundings outside the windows in the 3D world had a great influence on the spatial arrangements made by the different groups. All of them made substantial reconfigurations after they had visited the real-time 3D for the first time and become aware of the view through the windows. We also learned that the possibility to add accessories is important in developing the design as well as experiencing it.

Astrid Lindgren Child Hospital—Handling Negotiation

On a commission from Locum AB (a hospital real-estate company) we worked on the design of a new work environment for a care unit at Astrid Lindgren Child Hospital in Stockholm, the SABH group. SABH consists of 25 persons, including doctors, nurses and assistants, who work with advanced medical treatment of children in their own home environments. The workplace at the hospital is therefore more of a central unit and a meeting place for an ambulating team than an ordinary hospital treatment ward.

We conducted two workshops in January 2003, both times using FSD. In the second workshop the SABH staff worked collaboratively with Locum representatives in designing conceptual solutions with FSD for the new work environment. A challenge was that the users perceived the offered floor area as too small (250 m^2) for their needs. A lot of design effort therefore focused on handling and negotiating pre-formulated demands and efforts to fit into the given space. The groups succeed-



ed with this commission and produced three different solutions. These solutions were then boiled down to one consensus-based proposal.

The design of the real-time 3D environment was this time very soft in its appearance with watercolour textures and mild colours. In this application size and measures where crucial and a lot of effort was devoted to ensuring the exactness of furniture and other components. Also in this representation you could see the surroundings through the windows. Although the 3D representation was not at all realistic there was no comment or objection about that. The participants accepted that it was only a representation and we explained carefully that the task was to build ideas, not physical solutions. The real space was only five minutes away and the participants went there a couple of times during the workshop to verify our representation.

The participants reacted very positively to using FSD as a design and visualisation tool. In spite of the inherent antagonism between the users and the real-estate providers about whether the floor area was sufficient, the collaborative design work in cross-groups succeeded in building a sense of common goals and fellowship between the participants. The real-time 3D visualisations played a crucial role in this by providing a neutral ground for discussions.

UNDERSTANDING REAL-TIME 3D AS DESIGN ARTEFACTS

My endeavour to visually develop architectural real-time 3D visualisations aiming at supporting dialogue and creativity in collaborative design demands an understanding of how these visualisations perform in action. In the following I will discuss the way in which they can function as common design artefacts. In order to understand the complexity of visualisations, I will also propose a number of categories for mapping real-time 3D representations. Kathryn Henderson (1999) argues that visualisations, by referring to a multitude of codes on different levels, are extremely flexible in their interpretation and facilitates multiple readings. By doing so, visualisations establish a negotiation space where codified and un-codified knowledge can meet. Basil Bernstein (1971) introduces a distinction between restricted and elaborated codes. Restricted codes—such as the jargon or terms used within a profession or a work group—are understandable only to a limited group, generated by social structures and serve to transmit culture. They limit vocabulary and flexibility but at the same time point out assured group solidarity. The elaborated codes, on the other hand, are open, have explicit meaning, include unpredictable form, and allow for modifications to suit the viewer. Their function is to invite

different interpretations. In opposition to this dichotomy, Henderson (ibid) states that codes are embedded in visual representation and can be read on different levels by different viewers. A broad group of participants in a design process can be said to recognize code systems on different levels, neither elaborated nor restricted. A visualisation that contains embedded codes can be deciphered by those who understand the codes, but can also be interpreted on other levels, where associations and multiple readings form the common ground between different persons and parties. The richness of associations also opens up for the visualisations to act as boundary objects in the design process by inviting different participants to read different interpretations into the 3D worlds. This means that participants with different background and expectations associate different codes with the same visual appearance. In that way, by accepting it, the participants also relate to it and thereby achieve a common ground for dialogue and discussions.

Accordingly, for the purpose of understand visualisations as design artefacts, I think it can be fruitful to regard architectural visualisations as, intentionally or not, producers of meaning and references to internal and external codes. A visual expression can be considered as consisting of multiple different signs. A sign is interpreted by referring to a code; the code connects the sign to associations and understandings. The codes can be individual or collective, and links the visualisation to the external world. By starting to discuss some significant categories I will try to outline ways in which the signs are embedded and how they are connected to fields of reading and association. I think that Zobel's (1995) suggestion of six categories can form a good base for understanding. The concept of affordance is the basis for his effort to grasp how different media of representation (model, perspective, noninteractive screen-based animation, interactive screen-based animation and computer-generated immersive environments) describe the experiential qualities of space in architecture. These qualities are according to Zobel covered by the categories texture, form, colour, light, scale and movement. A visualisation communicates on many levels simultaneously, and the combination of the different categories makes up the total appearance. Each category, as well as clusters of categories, is tied to codes that can be read in multiple ways. All of them are not always relevant, but some of them can be useful in the understanding of what it is that constitutes a real-time 3D visualisation and how it communicates.

Texture: Texture refers to the fine structure of the environment at the scale of centimetres and millimetres. There are pigment textures and layout textures and they can exist on surfaces or in the environment as optical textures that determine the distance and size of objects in visual fields. Texturing refers to how different sur-

faces are shown in a representation, how the forms are rendered as material. The materials can have different looks and qualities in arrangement and reflection, be coarser or refined, etc. A representation can compensate low geometrical complexity with sophisticated texturing and in that way carry architectural information. Textures are extremely important as references to external codes. They are sign-intensive and comprise the core qualities in building metaphorical worlds.

Form: Is what physically makes up everything that can be touched and seen. Hilka Lehtonen (1993) examines how new visualisation techniques and the interest of the architects meets in visualisation. She makes an important distinction concerning architectural visualisations and the relation between model and representation. According to Lehtonen, model is the internal quality of the subject that is being modelled. Representation is the kind of presentation that is aimed for direct visual observation, which is the actual visual appearance. The form is the model in accordance with Lehtonen's definition, a »copy« of the architectural original. It has to structurally follow that geometrical prototype to be meaningful. A 3D model is, however, always a copy having a lower amount of geometrical complexity than the original. Between the real world—or a virtual prototype of a proposed building—and the 3D representation, there are different degrees of correspondence. Geometrical detailing belongs to the form category and refers to the number of surfaces and facets in the representation. The higher the level of detailing is, the more geometrically rich and complex is the representation. This also means that the representation grows in terms of information and polygons. The concept of detailing also covers the amount of elements and their complexity in a 3D representation. A 3D representation is said to have a high or low level of detail, which typically corresponds to density of information or level of complexity. A high-density representation is rich with information. A high level of detailing, however, does not always mean multiplicity of meaning and interpretation.

Colour: Colour is a property of an object that is carried to the photoreceptors of the eye by light. When mixed with surface texture, colour illustrates basic qualities of objects such as material and finish. The colour in a virtual representation can be applied as a texture but textures can also have different colours. Traditionally, colours are associated with expressions and emotional experience as opposed to shape and form that are linked to communication and intellectual experience (Arnheim 1956). Historically, colours are carriers of symbolic content. This symbolic association varies from different cultures, but they can still be very much alive as collective memories.

Light: Gives characteristics to surfaces and objects that help to understand forms and their particular surface layout. Light and shadow can be used to create and enhance space in geometrically simple representations. A sensible lighting of low-detail spatial representations could be one way to create them plain and simple. Still, they must be informative and have the possibility to communicate the architectural purposes and qualities. Light has the power to make spatial representations »come alive« without having to overload them with irrelevant details.

Scale: You can distinguish between three types of scale: physical, proportional and human. An interior space or a building is never situated in nowhere. The possibility to comprehend where you are in space and relate to other spaces and buildings is perceived as being fundamental. In real life you always refer to the outside through windows in a building. Persons, furniture and other equipment give the virtual space scale and meaning. You can relate to it and translate it to your everyday spatial experiences. In this category you can also include atmosphere-building objects such as animals, greenery and art.

Movement: The world is perceived in motion and not as a series of still images. Movement reveals different depth and scales. What distinguishes architecture from most other artefacts and means of artistic expression is that it limits and encompasses space. Buildings are hollow and you can enter them and walk around in their interiors. The experience of architecture (Rasmussen, 1957) is temporal, a process in space. The reality of space cannot be experienced from one single point. A visual representation in 3D displayed on a 2D screen therefore needs time and perspective representation to be experienced. When moving around in represented space you are able to experience an architectural visualisation spatially. In real-time 3D, the drama of architecture is performed at your own pace. In the western cultural hemisphere, central perspective is an established convention for representing space since the renaissance. According to Erwin Panowsky (1994), it symbolises the rationalisation of the subjective view into »science«, and its history can be seen as a »triumph for the distanced and objectifying sense of reality.« His point is that the central perspective is not a neutral way of representing space; it is a carrier of symbolic form. In spite of its objective ambitions it is obvious that the central perspective only represents space as perceived by a one eyed viewer. Computer made 3D views are calculated according to the laws of central perspective. If the perspective is constructed with a short distance (wide »field of view«), this creates a good sense of depth, but distorts the image in the periphery. It also creates an obtrusive close-

ness and sensation of the space. With a longer distance (narrow »field of view«) you create a distance that could get in the way of experiencing space.

VISUAL DEVELOPMENT IN THE PRESENTED PROJECTS

Throughout the series of described case studies, we have constantly developed our design of the 3D representations. They have evolved from a garage metaphor with no textures and coarse interiors to softly textured spaces with exterior views, accessories and populated with individual characters. This has been a process guided by intuition and based in previous experiences from architectural visualisations.

The transition from professional 3D in the Cave to game based visualisation of architecture meant several changes concerning the visual appearance of the real-time 3D. The input software you choose also determines the level of detail. Professional architectural software such as AutoCad, developed with the purpose to design complex buildings containing a lot of geometrical information, are able to produce 3D representations of high information density. In the Chemical Department project we therefore worked with a relatively polygon rich representation. Starting to work with Half-Life on the other hand, Worldcraft, which is an editor developed for modelling in Half-Life, gives a very coarse and low-detailed representation. A high level of geometrical complexity was replaced by a more simplified geometry. From the game engine we wanted to explore its possibility to create lighted environments very fast. Our assumption was that architectural space with low geometrical complexity could be made understandable and appreciated supported with light and shadows.

The first time Half-Life was utilised for visualisation, in the Telia Research and Vasakronan projects, we used a visual language close to Half-Life's own. We even used prefab components such as furniture from Half-Life's library. We also rather early tested to populate our environments. Unfortunately, we were also forced to acquire the population out of Half-Life's component library. That means that it was monsters or soldiers that populated our first architectural environments. One advantage with Half-Life is that movement is simulated as actually walking around illustrated with sound of footsteps and the way you »bump« into walls furniture etc. You are also able to run, jump, move sideways etc, in accordance with the needs in a computer game.

The design and use of texture has during the whole process been of crucial importance. In early phases of the architectural design process the knowledge of detailed geometry of the planned building is very low. Textures are therefore a way

to make simple forms understandable and possible to grasp architectonically. We have developed our use of textures in several steps from monochromatic surfaces to soft and visually complex textures. Referring to computer game real-time 3D, Walther (2002) argues that the spatial coherence of the computer game relies on rapid production of perspective drawings and textures. The textures in their turn rely on genre and the users' expectations on the projected representation. Textures are extremely important and can fluctuate between naturalness and stylised features. To be able to rapidly import or create our own textures has been an important quality in Half-Life/Worldcraft. In the Cave projects we worked with rather soft colours. In the initial Half-Life worlds, while leaning on the components at hand, the colours were bright and of high intensity. In the last project we have more turned to soft and diversified colours.

When we switched to ForeSite Designer, the 3D representations also changed. In the Interactive Institute and Chalmers Studios projects we, among other things, tested to use softer watercolour textures. We also started to develop our own component library with modular walls and furniture. Thoroughly modelled individual persons with pre-programmed behaviours were added as a new feature, aiming at giving scale and sense of place to the 3D worlds. An important step in the development of our 3D worlds was when we took the step out to »sharp« projects. Previously, we had been acting in contexts that, although we dealt with external partners and participants, were clearly research settings. In the Experimental Office and Astrid Lindgren Hospital projects we were dealing with spaces that were going to be built in immediate connection to our assignment and that of cause put extra effort on our tools and process. Here we intentionally tried to design the 3D worlds as composed wholes that were visually harmonized. That means that the detailing, although still very schematic, was raised to a higher degree of complexity, apparent in the adding of surroundings, a richer repertoire of components and focus on colour and textures. It was now also very important with accurateness concerning scale and measures.

DESIGNING REAL-TIME 3D AS DESIGN ARTEFACTS

As independent artefacts real-time 3D architectural visualisations can be studied, analysed and intentionally designed as carriers of addressed messages. The design of real-time 3D as design artefacts, made for the collaborative use of non-professionals, brings up visual demands that are different from making presentations of architectural proposals. An important issue is how to put real-time 3D into play in

the early phases of the design process. How should they be designed so that they stimulate dialogue and enhance the creativity of group design work? Based on our own experiences, supported with referred studies of literature, the following mementos can be formulated:

Real-time 3D representations used in early phases of collaborative architectural design should take into account the following.

- *Detailing*. Have an information density coordinated with the design phase.
- *Simplification*. Eliminate features of real world complexity—support creativity.
- *Appreciation*. Support a positive attitude among participants.
- *Confidence*. The representations must be correct and trustworthy.
- *Openness*. Have a conscious and explicit reference to the use of analogies and metaphors.
- *Leaving Traces*. It must be possible to influence and »mark« the world.
- *Sketchiness*. Be visually open—support multiple interpretations.

Detailing (not adding)

Real-time 3D used in early stage of a collaborative design process is not particularly dependent on explicitly naturalistic representations of the architectural space (Fröst and Warrén 2000). On the contrary, our studies have demonstrated that it is important for the design representations to be understandable, with as low level of detailing as possible. In early design phases you work with fundamental decisions concerning spatial ideas and layout. Decisions on colour, technical details and furnishing are to come later. It is therefore important to develop real-time 3D representations where the level of detail is coordinated with the phases of the architectural design process. The detailing of a real-time 3D representation can be very influential on the design discussion and the process. The more information a representation contains (more detailing), the more detailed discussion it generates (Brandt, 2001). Our experiences correspond to Goodfellow's (1996) argument that too »finished« representations may actually bring problems into a design discussion as the users often feel that they can only approve or reject a suggested design.

Simplification (reduction)

A real-time 3D representation is always a simplification of reality or a virtual model. By showing students different 3D representations of a university campus with varying levels of detail, Tomlinson and Holmes (2002) demonstrate that low levels

of detailing in real-time 3D walk-through simulation are effective in supporting understanding of spatial relationships in a represented design. The correspondence nevertheless has to be trustworthy, but the simplification itself seems to open up for modifiability. The workshops conducted at the Interactive Institute and Astrid Lindgren Child Hospital are both examples of how the representations are used as «things to think with« by providing a simplified version of the complex reality that was just next door. By providing a simplified representation of real world it is easier to let that representation function as a context for experiments and open up for a creative, team-building dialogue. Simplifications make it easier to ignore complications in the task at hand. This is already noted by Schön (1983) concerning the virtual world of drawings and their ability to stimulate design by eliminating features of real world situations, which might defeat or disrupt experiments in the architect's internal work.

Appreciation

A real-time 3D visualisation is not merely a technical design artefact or carrier of information. It also has visual qualities, which can be attractive for the participants. Associations connected to 3D environments can elicit negative, neutral or positive reactions among participants. Negative reactions means that you tend to push the image and its content away from you. Our experiences suggest that positive reactions to the visual appearance support positive collaboration in the workgroups. By identifying a number of signs in interior images made by students of architecture, which were also tested experimentally for positive and negative reactions, Lauthers and Sjöblom (1988) suggest a method of interpreting how different factors in architectural visualisations convey moods and atmosphere to external viewers. Factors that were found to mediate the message of the images most strongly were colour treatment, perspective, portrays of persons and the appearance of views onto the outside world.

In our series of case studies, we have intuitively developed our design of the 3D worlds in a direction similar to the conclusions offered by Lauthers and Sjöblom (ibid.). We have incorporated several of the signs that they identified as being perceived as positive. The worlds in ForeSite have evolved from garage-looking »space labs« with no textures and coarse interiors to softly textured spaces with views, accessories and populations of individual characters (with constant model features). The representations have progressed towards greater semantically richness and the positive interpretations has been the magnet. The users appreciate these environ-

ments more and show a positive attitude towards the design artefacts and the whole design commission.

Confidence

Confidence is promoted when representations are correct. The participants must have confidence in the representation. This is something entirely different from accepting simplifications or low level of detail. When you work with real problems in real projects the correspondence between representation and reality can be crucial for the credibility of the process itself. This holds irrespective of whether the reality is an existing building or a virtual model of a planned building (in the latter case, the verification merely takes place in another media). Measures, scale and correctness in performance are crucial in making the participants feel trust and engagement in the commission at hand. The role a representation will play in real projects is not always foreseeable. It can be anything from using the space as a »stage for actions« to actually verifying if it is possible to put another desk and shelf in a specific workroom. When the representation is used for verification, then everything has to be correct.

Openness

To look at a representation metaphorically opens up for the explicit communication with manifest code systems. You represent something »as-if.« The Chalmers workshop shows that the use of a very explicit visual metaphor (Mondrian-style) was a way in that specific setting of making the very stylised representation acceptable. In the Telia Research project we deliberately emphasized the (unavoidable) computer game metaphor and in the Astrid Lindgren Hospital designed an environment that referred to a watercolour drawing. Instead of entirely being grounded in authentic experiences of reality and the environment our vision is often blurred with images from different media (TV, computers, internet etc), art and commercials. Gombrich (1960) sees meaning and interpretations of visual representations as governed by different culture-bound conventions. As they are dependent on culture they are therefore unstable and change over time and situation. The interpretations of certain visual manners are accordingly changing all the time. Our world has in many ways been turned into a world of signs. References and metaphors are consequently often used as a way to widen the interpretation of a visualisation. Rowe (1987) points out that in the development of initial design proposals, ideas are often appropriated from outside the context of a specific problem. An analogy is made with

objects and organisational concepts that are outside architecture. Schön (1983) identifies the use of »generative metaphors« and the procedure of seeing a design problem as something else (seeing-as) as playing a crucial role in invention and design. By letting the real-time 3D world consciously relate to an analogy or metaphor, for example a certain style or a game world , it can in that way open up to a whole group of codes and interpretations. If visual metaphors or references are used they benefit from being openly accounted for. This also includes references to the styles and types of visualisations.

Leaving Traces

When a real-time 3D visualisation is created and adjusted through group interaction, it might work as team builder in a diverse group of participants. My experiences show this effect in most of our carried through projects. An important quality in this respect is the possibility to »leave traces« in the representation. In FSD you leave traces in several ways: by actually building and manipulating the artefact, by having the possibility to leave messages inside the real-time 3D world (frequently used) and by ensuring that every version of the work is automatically saved. The representation is the output, created through collaborative effort on the spot. It immediately gives feedback on what has been achieved. This is also important from the aspect of maintaining achieved results and avoiding repetition of work. My observations are consistent with Brandt's (2001) conclusions on the importance of being able to »mark« representations in collaborative design work.

Sketchiness

A challenge is to design real-time 3D environments that function as sketches in collaborative design. It is important that a visualisation is open and affords multiple interpretations. How finished a representation appears, its openness, must be at such a level that the participants still feel invited to be active and contribute ideas. They must feel that they are dealing with a sketch and not a final presentation. Presentations rely on established codes and formalised manners in order to direct the field of associations into coherence and concordance. Sketches are characterised by being open and unfinished. According to Jurij Lotman, it is in the breaking with the expected that the visual effect emerges. Information is similar to abolishing something indefinite; a lack of knowledge is replaced by new knowledge. Where there is no lack of knowledge there is no lack of information. The total amount of alternative information consequently depends on the presence of complementary

possibilities. Information is the opposite of automation; if an appearance automatically results in another appearance, then there is no information. In this sense you can view the transition from a sketch, with a lesser degree of similarity, to a realistic rendering as a reduction of information inherent in the image (Lotman, 1973). Accordingly, a sketch has greater possibilities to contain information than a photo realistic rendering. Sketches are very powerful in terms of being semantically »deep« and communicating on different levels. Finally, returning to Henderson (ibid.), she sees the quality in sketches in that they can freely jump between contexts and different grammars, produce meaning on many different levels and in that way refer to a multitude of codes. They can build on well-known examples, refer to established knowledge and codes and juxtapose it with new invented components or using metaphors from popular culture, comics or whatever.

> It is as if in verbal communication one could speak or write using the vocabulary, grammar and syntax of Standard English, work in some French and Russian phrases as well as some newly coined words, phrases, and grammatical structures; and then expect monolingual English speakers to follow the meaning. (Henderson, 1999).

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HEROINES, SATYRES AND SACRED WOODS: NOTES ON NARRATIVITY IN INTERACTION DESIGN

Ylva Gislén

PROLOGUE: WHY NARRATIVITY?

The last decade's exponential growth of academic literature on narrativity is an interesting phenomenon in itself. Here is not the place to give a complete overview and discuss how the topic have been and is treated in as different disciplines as linguistics, sociology, psychology, cultural studies, law, theology, gender studies etc. But a few remarks about some commonalities are necessary in order to understand the possible relevance and importance of narrativity to interaction design.

The interest in narrativity seems to correlate with a general move towards a view on identities and communities as socially constructed. The idea that these entities are not once and for all fixed, but exists as temporary compromises between historical continuities and permanent change are tightly linked to the notion of reality itself as something that we describe and handle primarily through the arbitrary means of language. From the Wittgensteinan approach to the relationship between language and reality, and from the Foucaultian notion of discourse as something that produces both statements and a certain range of possible subject positions, the step to the hypothesis that narratives are constitutive means by which we negotiate meaning and causality has indeed not been a very long one to take.

The interest in narratives and narrativity in the abovementioned academic disciplines are expressed as attempts to examine how, generally and specifically, these processes work, with an emphasis on the relationship between common and individual narratives, on the relationship between elements of construction and reconstruction.

In interaction design concepts and understandings of narrativity could possibly enter the arena in a number of ways. As I see it there are three levels, or areas, in which narrativity are important to consider.

The first one is, evidently, the area of interactive narrative. The digital media have given birth to a number of new narrative formats—and the debate over in which ways interactivity affects the different means of narration and the respective roles of reader and author has been intense. Many interaction designers are and will be working with designing computer games, hypertexts, autonomous characters, multi-user domains etcetera, and are and will be dealing with issues of narrativity very explicitly.

Another area is the use of narrative in the design of interaction in general. Already in the eighties Brenda Laurel wrote *Computers as Theatre*, and suggested drama as a model for understanding and creating better interfaces and interactions. I will discuss the strength and weaknesses of her strict aristotelian model below, so far it suffices to remember that she, and others after her, have pointed to the possibility of using narrative as means of understanding and creating interaction itself.

The third area is the largest one. If Klaus Krippendorff is right in stating that the design dicipline itself is now moving on a trajectory of artificality, going from artefact to discourse, and in suggesting that the new prime design principle should be »meaning is the only reality that matters« (Krippendorff, 1997), then narrative, as one of the prime ways human beings organize reality in a meaningful way, certainly is very important to us as designers. Both in regard to how we set up and think about design processes, as well as in the ways our products and concepts enter and change spheres of human life. I am pointing out this third area, well aware that every attempt to grasp what a narrative approach could mean in this enlargened sense, risks being very vague.

Still I will, in this article, attempt to move between these different areas, or rather possible perspectives, on narrativity in interaction design. I will do so primarily in the company of Umberto Eco's Norton Lectures: *Six Walks in the Fictional Woods*. This because he, in the condensed form of the Norton Lectures, discusses many issues and concepts that I think are relevant also to narrative in digital media.

I will thus start with a basic discussion about different models and understandings of narrative structures, continue with a reflection on more particular concepts and notions that I have found to be useful, and finish with a discussion about how these narrative concepts can be related to a social and political awareness of the designer.

I. DOES OUR UNDERSTANDING OF NARRATIVE MATTER?

Preceeding the discussion about what difference interactive technologies and media do to the concept of narrativity was a slight shift of focus within narratology , literature and theatre studies: from the artists and the narratives themselves towards an understanding of narrative that incorporates—or recognizes—the role of the reader (or the audience). Roland Barthes claimed that the text is actually not »created« before it meets the imagination and understanding of the reader (Barthes 1977). Umberto Eco describes the »Model Author« and the »Model Reader« as two entities that mutually create each other, not only in the form of textual instructions and explicit narrative strategies, but also in the form of potential and actual assumptions and expectations (Eco, 1994).

This shift of focus was, at least partially, provoked by the destabilisation of the concept of the essential meaning of a certain text throughout different contexts, connected to the intentions of the empirical authors. When the homogenity of the western culture and the western reader (white, male and well-educated) dissolved into discontinuities and heterogenities it maybe became easier to see the meaning of the text as an interplay between context and text, between author and reader.

Many of the thoughts and concepts from this reader-centered narratology have been brought forward into the discussion of interactive narrative, such as »open work« (Eco, 1989/1962) and freedom for the reader. What tends to be forgotten is that even though these theorists have questioned and opened up the meaning of the text or narrative, their thoughts are still very tightly linked to a notion of the text, be it a novel, a film or something else, as something stable and fixed in its material aspects. I think this is what Espen Aarseth is trying to point out in *Cybertext*, when he claims that metaphors such as »worlds« and »labyrinths«, used by these contemporary literary critics are misleading and would have more analytical accuracy when applied to texts that are like that in a more literal sense—nota bene what he calls ergodic texts (Aarseth, 1997). I do understand his attempt to distinguish between the meaning-making involved in every reading from actual, material change of and interaction with a text. But he misses the point of those concepts being meta-

phors. In a metaphorical sense it is very useful to look at every text, be it interactive or not, as something that opens a world, a forest or a labyrinth: a sort of a common ground which the author and reader collaborates in creating. Not doing so would be to dismiss the most powerful tools in creating and navigating narratives we have, namely our expectations and imagination. This is important to keep in mind, since we as interaction designers enter the arena of narrativity and digital media with a focus on the user's actions and experience, which requires a different approach than that needed in making descriptions and taxonomies.

I will develop the question of expectations later, and return to the question of the materially fixed text. It is an important one, not because the structural divide between media like for example the printed book, the film and the many and diverse formats in digital media necessarily depends only on this (or is that sharp), but because our current understanding of narratives is tightly linked to the paradigm of the fixed text and to a certain relationship between author, text and audience, in a way that I believe hinders our understanding of the different possible narrative formats and structures in digital media.

Begin with the concept of text—which I admit I've here used purposedly in a sligthly ambivalent manner, although designating the whole corpus of material narrative discourse—be that a performance, a novel, a film or a composition. Only for a very short time in the history of the human being has narrative been tied to the written word. For a large part of our history, and still in many parts of the world, the act of narrating was and is closely linked not only to the spoken (sung) word, but to music, rhythm, gesture as well. The ancient Greeks, whose innovations in dramatic narrative have had such a large impact on western storytelling, would propably feel more at home in a contemporary rock concert than they would in any contemporary theatre building. Not only because of the musical and gestural orchestration of the lyrics, but also because the audience sing and move along in the former.

Every historical and geographical attempt to understand narrativity, that tries to stretch out beyond the boundaries of the last hundred years of western civilization, will soon also result in a blurring, or transforming of connotations to categories such as audience/spectator/reader. However wildly participating in our imagination, we seldom connect these categories to real participation and/or interaction, and also tend to see the narrative experience as an individual experience. But how to frame the different positions in an African call and tell performance? Are the categories at all useful in describing the praxis of aboriginal songlines where every persons singing and walking the land is a vital contribution to keeping it alive? Can we be sure, even when turning to the Western tradition, that these concepts accu-

rately describe the experience of the different levels of participation in a medieval mystery play, or of the participants in the dionysic rituals where the greek tragedy took place, or for that matter in a contemporary role-playing game?

The research into the diverse field of what is called oral tradition and performance have so far also moved, from a view on narrative as something that before the written word was memorised and repeated to something that is composed in the moment, in interaction with listeners and participators, out of a communal repertoire of storylines, characters and, at least in the epic formats we know from Balcan and North Africa, what is called formulas, more or less stable expressions (Lord, 1960; Ong, 1982). This does not exclude, as Ruth Finnegan has pointed out, that there are places and times for the precomposed, through rehearsal, fixed text in oral tradition and performance (Finnegan, 1988), but her remarks do not change the fact that the picture of »text« or »the artistic narrative piece«, as something that must be a completed and fixed whole, no longer is the only possible one. Narratives in this sense do not necessarily have any material representation or form between the moments of performance, and continue to change in relation to a changing social context, as well as that there can co-exist several versions of what still can be seen as the same storyline. The idea that a text could exist independently from an audience is also foreign to the oral tradition (Ong, 1982).

And then, what conclusions can be drawn from this loosening up and change in perspective on what constitutes a narrative and the different positions we associate with the act of narrating and performing? I think there are basically two, both important to the field of interaction design.

The first one is that there are a large number of narrative and dramatic structures, formats and practices, many of them involving participation, interaction and change, that can be tried, reformulated and used when creating narratives for digital media. The same freedom, (although that freedom is somewhat moderated and restricted by the possible agreements we can enter, which I will discuss later) , the same diversity of repertoire also counts for modelling interaction narratively.

Brenda Laurel (1993) made an important contribution to interaction design when she introduced the thought of shaping interaction in dramatic terms in *Computers as Theatre*. Unfortunately, as Pamela Jennings have pointed out (Jennings, 1996) she limits herself to a strictly prescriptive version of the Aristotelian model: with protagonist, antagonist and a rising curve of dramatic tension, towards a climax and a resolution. Restricting ourselves to this Aristotelian model is problematic for several reasons. First, there is no evidence for it to be culturally universal—which makes it questionable as the only possible model. Secondly, a reference to the Aristotelian model does not come with any incorporated practice of interac-

tion—and putting the user in the position of the protagonist or the antagonist is neither an obvious, nor an uncomplicated choice, as we will see later.

Thirdly, the rising curve of dramatic tension is not very helpful when it comes to modelling interaction over time—which we often associate with a narrative approach to digital media or interaction design. (Finally, one can raise some doubts about it for both historical and political reasons. I will come to that in the last passage of this essay.)

Why is it so? Because, and this brings us to the second possible conclusion from, or rather elaboration of, the discussion above: the perception of story and storytime seems to change the moment we move from the position of spectator and reader to performers and interactors. The first time I got a clue about this was in the middle of the nineties in Sweden, when there was a virtual boom of standup comedians, theatre sport, and improvisational theatre events. With a lot of curiosity and fascination I witnessed the engagement and participation of the audience, and how what, from a more distanced viewpoint I sometimes took as a theatre reviewer, could be called more mediocre or inconsistent stories, apparently were perceived in a different way by spectators who were invited to comment, propose elaborations and possible outcomes of the events on stage.

When studying role-playing games, as a contemporary practice of interactive storytelling, through interviewing gamers and game-leaders, my observations were similar. Although many of the gamers and game-masters referred to the aristotelian time-curve, and stressed the importance of causality and closure, they also expressed the importance of a great flexibility and tolerance to in which order events take place, and for how long they are elaborated on—as long as the story experience, in the actual moment of it taking place, is perceived as satisfactory (Gislén, 1999).

In the experiments made with creating interactive narrative environments in the Oz project at Carnegie Mellon University the conclusions were similar. The dramatic experience, especially concerning time, were, to the researchers' great surprise, perceived very differently from the different viewpoints of interactors and audience (Kelso et al., 1993).

I will not claim both time and timing are not essential and necessary properties of a narrative, but merely argue for a certain cautiousness in transferring criteria from author/audience positions into the interactive experience, or to restrict ourselves to the most simplistic models.

But one could also make some other reflections.

One of the interesting things about these traditional and new forms of storytelling and drama is that they contain or reintroduce a vital element from children's

pretend play in them, namely a continuous negotiation of the fictional elements and the course of events, as well as about the power balance between the different actors in the creation of a common fictional universe. What becomes visible are the pluralities and complexities that make narrative such a necessary and challenging way of meaning-making. Each story contains, as being a pathway through a more or less fictional space, a multitude of other potential storylines or deviations—that may or may not be instantiated. Umberto Eco describes this very lovingly in *Six Walks in the Fictional Woods* as the »lingering in the woods«: the inferential and imaginative walks outside the actual storyline that the text invites the reader to do in order to predict and guess what is going to happen, based on how the fictional space is described and based on her experience from reality and previous stories. And sometimes the »reader's« inferential walks will become realized as new stories, as the vast phenomena of fan-fiction, further facilitated by the internet, shows: new narratives starring the characters from often well-known contemporary stories, and in the setting created by the original author, but with often surprising turns and twists (see, for example, http://www.fanfiction.net).

These inferential walks are, as already mentioned, one of the more powerful tools the author of a novel, a drama or a film have in order to create the hopes, fears and tension attached to the emotional narrative experience—a skilled author predicts these walks and plays with them in order to make the pathway she has chosen interesting and surprising.

In a highly interactive narrative form this could pose some challenges, as many of the role-playing game leader I have interviewed stress: the players will always surprise you, and find tracks you haven't been able to foresee. And the players on the other hand point out that finding these tracks and thus extend the common fictional space is a vital part of the pleasures in role-playing, also in that it contains an element of testing the skill of the game-leader, as a constant negotiation not only of the fictional agreement, but also of the more social agreement of being in the respective position of storyteller and inter/actor (Gislén, 1999).

This leads us over to the next passage of this essay:

II. THE WILLING SUSPENSION OF DISBELIEF

That tries to deal with the inevitable question of where we can start then, as interaction designers dealing with narrativity? We do not have to bother about narrativity disappearing with interactivity—interactivity with a computer system might be new in the human history but interaction between storytellers and audience, between

narrative and context is rather the rule than the exception in that same history. In such a perspective the hairsplitting discussions on the difference between non-linear and linear narrative no longer seem relevant either. But this might instead give room to other anxieties. If we have such a large range of possible narrative formats and discourse to explore and relate to, not only from the realm of oral tradition but also from the last centuries of innovations in filmic, literary, musical and dramatic discourse, in comics and role-playing (not to mention what have already happened so far in computer and online gaming, in MUDs, avatar worlds, etc.)—is there anything steady or is the design of narrative or the narrative design of interaction also something that melts into air?

It's daring even to try answering that question with anything else than a confirmation, but still I will try, by sharing some reflections on some concepts. None of them are new ends of the nested threads of Ariadne (who said that clever woman only had one?) through Daedalus' labyrinth, but I have so far found them useful in trying to understand my own and others design experiments, be they directly focused on narrative or not.

Agreements—be conventional and prepare to negotiate

Entering a narrative—or to start playing along with the rules in narrative discourse—is , as Umberto Eco puts it in *Six Walks in the Fictional Woods*, »a tacit acception of a fictional agreement« (Eco 1994). I deliberately reproduce the word fictional, even though narrative does not equal fiction. But, as Eco explains and demonstrates, the line between fiction and reality is not that clearcut: fictional worlds are, as he says, parasites on the real world in that they rely on readers' assumptions about some things to be more reasonable than others. The way we accept the representation of the actual world, on the other hand, scarcely differs from the way we accept representations of fictional worlds. And, as Jerome Bruner has shown, we tend to interpret our everyday life in the form of stories (Bruner, 1990).

Be the narrative realistic or fictional: what matters is that we, audience/reader/interactor/user and author/designer cooperate in constructing a universe possessing a kind of internal cohesion (Eco, 1994). This cooperation requires an agreement—very similar to when children with the help of often subtle signals move back and forth between the modes of pretend play and »reality.«

Our life with narratives is full of those tacit agreements, both on how we should act physically during the act of a narrative, and on in which way we participate with

our imaginative powers. We know what to do with the chairs in a cinema or in a theatre building, we now that we are supposed to turn the pages in a book one by one starting by the first page. These are agreements that have become so self-evident and strong that they have become conventions—in a neutral sense of the word. This also goes for cutting techniques in the narrative discourse of filmic language, acting methods on the theatrical stage, etc. We know how to think, how to cooperate in directing our attention. The positive thing with those kinds of agreements becoming conventions is that they allow for both a concentration on the more immaterial aspects of the narative experience—and a mutual play with them. As Eco says à propos literature: avant garde narrators wants to upset the expectations of the reader, and model a reader who expect complete freedom of choice. But they can do so, and this freedom can be enjoyed, because the tradition of comprised narratives, because there are assumptions about some choices more reasonable than others (Eco, 1994).

The drawback is that the agreement over time can be so self-evident that it loses its strength. And many experiments in the narrative arts during the last century have been about renegotiating these agreements and building newer, momentarily stronger, ones.

When it comes to interactive media, and creating the conditions for what Coleridge called a willing suspension of disbelief: an agreement on participating in a narrative or participating in the use of a narrative discourse—we build on already existing conventions, but sometimes have to create completely new ones. The more unusal an interaction or a situation—the more we have to allow for an exploratory phase, a finding out about which are the material and immaterial rules. A playing with the rules is not really possible before they are mutually established—which is why the cries about a Shakespeare being impossible in interactive media are as pointless as asking a three-month baby to walk on her own.

This also bears on the ideal of immersion—expressed as an ideal of smooth, unnoticed interaction and an as realistic representation as possible. In fact the more energy invested in making the narrative agreement, the more participation in the creation of the inner logic of the narrative discourse, the more participatory imagination required—the stronger is potentially the narrative experience. But the negotiation process of entering an agreement, and keeping it, is a delicate matter.

Positions—and restrictions

In *Six Walks in the Fictional Woods* Umberto Eco discusses the position of the reader in literary fiction with the help of the concept of Model Reader. He shows

how for example James Joyce's *Finnegans Wake* foresees, and even requires a Model Reader with infinite encyclopedic competence, while Alexandre Dumas' novels would be very irritating and disturbing for a reader who checks the ancient maps of Paris during the reading.

Despite the fact that the possibility of interactive narrative in digital media have been around for several decades now, and despite the fact that the discussions often focus on what consequences a more participatory role of the »reader« have on narratives, there are strangely enough seldom a discussion about in which position the »reader« of an interactive narrative enters or participates in it. This is strange because one's position in or versus a story is one of the most important creators of expectations, and thus also an important part of the agreement. My expectations on possible actions will be entirely different whether I think I am going to be the heroine, her lover, her bitter enemy or her servant in the drama—or if I think that I am going to be the God-author guiding them all, or the composer, or maybe a commentator. This counts for all sorts of narratives in terms of identification, but it becomes even more important when you are supposed to act and interact.

I have found A J Greimas' actant-model useful in trying to think about different possible positions. (Greimas, 1966). It is also a very simplistic model—as you soon would find out trying to do an actant-model of any of Shakespeare's plays (ending up with several subdivisional actant-schemas), but at least it allows for slightly more sophisticated ways of thinking about positions than the far more limited protagonist-antagonist-models, or the use of a first-person/third-person perspective. What's important in this structuralist theory of narratology is that to each position can be assigned a specific action-sphere in relation to the overarching conflict. In interaction design this will parallell the possible interactions and influence you as a designer create for the user—a challenge including ways of building appropriate expectations on that same action and influence.

We can as example take one of Chris Hales' hypermovies, which all are extremly effective in shaping the user's interaction in relation to a position vis-a-vis the story. In *Jinxed!* a man is trying to get to a job interview. All the interactional possibilites you have as a user have to do with causing him trouble to get there in time. You can delay his alarm-clock, make the soap fall from the sink and causing him to slip, etc. The possibilites of actions are all rational in relation to the position that has been assigned to you—the resistant of the subject's wants, which makes this short story a quite satisfactory experience—in fact I would claim that its very narrativeness comes from this thoughtful way of restraining your possibilities of interaction, where making random objects clickable would have exhausted it.

The actant-model also shows why the position of the subject in the overarching story can be slightly problematic, since we in general make assumptions about the subject position having a quite large action sphere, and maybe even being able to alter the driving conflict. If the position is not restricted by other means, as for example very strong genre conventions (which is propably why shoot 'em up-games successfully can put the user in a subject position—this said without claiming that those games necessary equal narratives) it can lead to severe frustration on the part of the user. This doesn't exclude it as a possibility—but it urges us to be cautious.

In analogue role-playing games, where the players usually are assigned to a sort of collective subject position in relation to the conflict, much effort is put in modelling the characters so that the game leader can be sure in counting on the players to act consistently throughout the narrative. And in live action role-playing games, with up to a thousand participants and the gameleaders having very restricted possibilities of influencing the many courses of action, a large amount of preparatory work is put into the collaborative process of deciding and making clear in which positions the different participants are in regard to the general conflict (Gislén, 1999).

This is not contradictory to the fact that there are subject aspects of every actant—in a dramatic performance every actor on stage perform both out of a subjective understanding of her character's wants and wishes, as well as from a more »objective« understanding about what is the overarching story. (And this ambivalence, this spatial simultaneity, is in my view actually what makes drama such an interesting narrative format.)

There are also other possible positions of the user »outside« the conflict described in the actant-model, such as the creator of stories, such as being a commentator or a revealer of the conflict (as in a detective story, which together with the variant, where the subject suffers from total amnesia, seems to be transformed quite easily to the computer-game format). One can also think of deliberately changing the positions of the user throughout a narrative. And when you are dealing with multi-user media the issue can potentially become very complex. The important thing to remember is that position—the relation of the user vis-a-vis the story or the conflict—is crucial in both narrative and interaction design.

Subtle or complex interactions

When starting to think about and investigating interactivity in relation to narrative, many of the already existing theories, especially concerning hypertext, equated moments of interaction with choices, or articulated deviations. But when interviewing

role-playing gamers my perspective changed. They stressed on the contrary that they avoid explicit choices between two or three alternatives, and that the actions the players performed seldom influenced the fabula (if the game-leader had designed the session with such a scenario—many of them simply create a conflict and let the narrative evolve from that). The fabula meaning the basic storyline—the player's actions are definitely otherwise essential to the whole experience. The feeling of participation and co-creation of the narrative experience was, as they expressed it, more important than the degree of actual influence over the story, and definitely more important than a clarity about choices and possibilities.

Some of them draw an analogy to how you perceive your own life, where you very rarely know when you are at important decision points: which actions will be influencing the future course of your life the most, and in what ways.

Such an awareness of the complexities and difficulties in rational decision making echoes the claims Ulrich Beck makes about how people perceive both their own life and public decision making in the context of a contemporary world which seems full of potential risk and hazards. Beck writes against the backdrop of the environmental crisis, that during the last decade have been growing from unexpected seeds in the middle of the once so orderly arranged garden of rational progress. But he extend his argument about this uncertainty to almost all the areas of contemporary human life: in the areas of social relations, of work and education, of health and disease. The promise of the Enlightment was that the more knowledge we would have about the human and non-human world, the more certainty we would achieve. Instead the result has been an ever-increasing uncertainty, an awareness of the hitherto unknown risks that could evolve from our actions, even from whithin the nucleus of our cells with potential hereditary factors beyond our control (Beck, 1992).

I have the impression that the interest in the combination of interactivity and narrativity reflects this anxiety: could there be places where we in symbolic forms can picture and handle these notions of risks and complexities, of stories with no secure endings? And do not then the alternatives presented in the forms of formats of explicit choices and a great deal of influence for the user address this anxiety at the entirely wrong plane?

Here, I believe, lies the potential for a different and possibly fruitful approach for both interaction design and narrative in the domain of digital media. With a shift of focus from the clarity and functionality in interaction (which, I would like to stress, are not values and properties that we should abandon altogether) to the importance of participation, co-creation and incorporation something interesting might happen.

An example that might clarify my point: there have been some experiments with interactive theatre and film where the audience have been provided with sets of green and red buttons, that at some points in the performance have been used to decide, with a majority vote, whether the story should evolve in one or the other of two possible directions. These turn out not to be particularly interesting or involving experiences. Compare with theatre performances where skilled actors, just through noticing and incorporating a laughter, a cell-phone signal or an odd movement somewhere in the audience in what's happening on stage, creates the feeling that the different individuals in the audience matter, that they are recognized and that the story wouldn't go on, or look exactly the same, without them. Those kinds of subtle interactions or complex feedbacks, which are the sum of many parameters and several people's actions, are in fact much more rewarding and adequate in the realm of narrative experience. And thus creating common grounds for dealing with complexities and participation could prove more valuable to us all than trying to streamline meaning, claiming freedom, clarifying choices.

Spatial relations

The notion of space is important in narrative, even if it is purely text-based. Umberto Eco refers to the rethorical figure hypotyposis: the lingering of the text so that we »see things.« This notion of space is often connected to the pace the text imposes on the reader: if the discourse time and the reading time is extended in relation to the story time, the impression of space will be more pronounced. In the art of space, nota bene architecture, this equals circumnavigational time, where rich decoration and scale represents a certain imposition on walkers and viewers (Eco, 1994).

Virtual spaces and physical interfaces offer new possibilites of using spatial rethorics in narrative—as well as constraints. (There are for example some problematics connected to the »materiality« of the virtual space, as imagining the space is one of the areas where a conventional reader puts much of her energy and potentially retrieves a large amount of pleasure. In role-playing games events are organized spatially—but with the advantage of the location of them being flexible, since the space is an imaginary one).

The particularity of the spatiality of narratives in digital media is something that has puzzled me more than other things, especially since I amm most concerned with narrativity in spaces occupied by several users—be they physical or virtual. I have asked myself whether the fictional landscape in itself is the actual starting point for storytelling—much as I remember myself as a kid together with my friends creating fictional landscapes in the woods. Sometimes they were miniature land-

scapes with castles made out of sticks and leaves—sometimes they were a common fantasy in which the old beech became a mountain and a pile of stones a witch's castle. In either case they became the containers and triggers of sometimes month-long pretend-play stories. In *Mindscapes: Geographies of Imagined Worlds*, one of the authors, Frank McConnell, even claims that the creation of fictional worlds is preceding the narratives: that it is the possibility of a space with new and other rules that conditions storytelling (McConnell, 1989).

But the primacy of space in narrative was for me contradicted by the observation that although much effort in MUDs and avatar worlds is put into creating spectacular spaces, they do very rarely become rich in narrative. It might have to do with the fact that in many of these social places the possibility of creating a personal space is used as one of the prime means of creating individual expression—which works against the essential common nature of narrative. But the Canadian dramatist and director Keith Johnstone gives a sligthly different perspective. In introducing the art of telling in front of an audience he tries to come up with a definition of story that matches his own praxis of improvisational storytelling.

As necessary as the flow of free assocation, the mere telling of events might be, he points to the cruciality of reincorporation: that story happens when the different objects, actors, places in the telling are reincorporated, or put in a significant relationship to each other. Temporary closure happens when the pattern of significant relationships is completed (Johnstone, 1985). This might sound somewhat mystical—but I would still like to claim that starting to handle the specific properties of spatiality in digital narrativity has to do with figuring out how to load these spaces with internal and symbolic relationships to their inhabitants

Oral tradition offers a rich inspirational source when it comes to spatial organisation of narrative: the memory-palaces in rhetorics, the episodical epic (like the *Odyssey*), or the already mentioned aboriginal songlines, where stories are laid out in the real landscape, to mention a few—but in all these cases the issue is not so much about using architecture or geographies as backdrops for stories, but about loading real or fictional spaces with significant relations, establishing webs of meaning.

III. AESTHETICS, ETHICS OR WHY THE TRAGEDY OF THE COMMON NEVER SHOULD BE END OF STORY

I think we must be aware that in all the talk about narrativity lies hidden the danger of essentialism: a claim that some existing narrative structures, like for example

the Aristotelian model, or the heroic myth, are essentially true or »natural.« In *Six Walks in the Fictional Woods* Umberto Eco refers to our tendency to interpret what happens to us in terms of what Roland Barthes called a »text lisible«—that we try to read life as if it were a piece of fiction. He continues to refer to Jerome Bruner and his theories that our understanding of everyday experience takes the form of stories, and to the theory of case grammar: that we understand sentences because we imagine short stories.

Ecos' aim is to discuss the relationship between fiction and reality—and for any lover of drama and story it is tempting to assign such a power to narrative structures. But combined with claims that for example the heroic epic is in some way intrinsic and unavoidable in the way human beings understands and act in the world (Campbell, 1949) is to fall in this essentialist trap.

Even the actant-model, which I for now found so useful, comes with such universalist claims: not only that narrativity is the organizing principle of all discourse, but also that the actant model is the one and only possible narrative skeleton (Greimas, 1966).

Why should we be aware of this as a danger?

Because if there is such a relationship between narrative and reality, then the choices of narrative models certainly aren't innocent matters, and every universalist claim should be regarded with suspicion and skepticism.

Take for example the Greek tragedy, that through Aristotle, who described and reflected over what he saw on the Greek stage at the end of the rather short period during which the Greek drama was invented, has survived as the prime model for how a narrative should be constructed, via the help of the french-classicists (the renaissance dramatists did not take any rules by the book) the nineteenth-century theorist Freytag and the late Hollywood cinema.

Even if we for some resons would like to claim that the Greek drama should be the model—how can we be sure that what we picked up resembles the experiences of the ancient Greek audience? From what we know each dionysic performance was divided in four parts, with three tragedies played in a row, and concluded with a satyric play. If we for one moment consider this as narrative structure instead, it consists of a cyclical repetition of three tragic stories, ended by a loose-ended turning upside down of this tragic order in the satyrical play. There are also some etymological evidence that the division between tragedy and satyrical play wasn't that sharp in the beginning, since the word tragedy means »song of the malegoats.« Just out of this small historical excursion the model of the Greek tragedy turns out to be much more haphazardous than is usually claimed. But what have been the consequences of a narrative model that focus on a collective catharsis through death and

disaster? What are the implications of the Hollywoodian version, that through the cathartic death and disaster almost always in some miraculous ways arrives at an happy ending for the hero?

Donna Haraway shows, in *The Cyborg Manifesto*, how the story of the Great Progress and the final arrival to Utopia in the Western tradition is tightly connected to the idea of a Harmageddon, and how different political movements, from the revolutionary worker's movement to early feminism tend to reproduce these fatal narrative structures, just replacing the hero with a new one (Haraway, 1991). I read her text partly as an urge to question, not who are assigned as heroes, but the meta-narrative structures themselves.

(Not to avoid the questions of death and disaster and conflicts in narratives, since the strength of narratives are that they offer a possibility to, at least physically, safely deal with these issues.)

But as both Donna Haraway and Michel Foucault show that does not mean we can create entirely new narrative structures—or discourses—from scratch (Haraway, 1991; Foucault, 1993). What we can do is to be aware of which kind of narrative models we do build upon, which discursive subject positions we allow for ourselves and others, how we formulate conflicts and conditions for engagement—to at least not be innocent about our aesthetic choices.

This non-innocence applies for all the three areas mentioned in the prologue. The example may be blunt, but if we, as designers, see ourselves as the heroes that are going to save the user-princess in the tower (not even questioning if she wants to be saved or not), or if we see ourselves as the humble helpers of the user-hero, who already has quite a good picture of what he wants to achieve, or if we see ourselves as the joint subject, fighting the monster of capitalism, or if we see ourselves as Arlecchino in a perpetual struggle with Brigella, figthing over the definitions, worth and use of objects, as often silly as innovative by the spur of the moment—will this not make a difference in terms of agency?

Umberto Eco also makes an interesting reflection that could serve as a warning against any overly strict application of narrative models, when he states that cult phenomena in literature—when fictitous characters start to migrate in to the real world and stories and their worlds take on a life of their own—seem to coincide with a disjointedness of the work. »In order to become a Sacred Wood, a wood must be tangled and twisted like the forests of the Druids, and not orderly like a French Garden.« (Eco, 1994, p. 128)

Another reflection Umberto Eco returns to and varies throughout the six essays in his book, and by which I will finish my own, is the shared nature of narrative. A narrative is, as he says, not a private garden. It is a place where the individual and

the collective memories and experiences meet. And in times when we more than ever need, as Anthony Giddens (1994) writes, discursive spaces where tradition and cultural differences can be negotiated. We will also thus continue to need rich narrative spaces: spaces where the narrative elements and the courses of events can be negotiated, where conflicts can be formulated and dissolved, meanings multiply and overlap. If that is the hidden promise of interactivity and narrativity, then the overarching question is whether we, armed with non-innocense, humour and irony, can create those possibilites in digital media and interaction design?

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A VIRTUAL REALIST PRIMER TO VIRTUAL WORLD DESIGN

Mikael Jakobsson

PLACES IN CYBERSPACE

Metaphors play an important role in our understanding of the world. They can both broaden our conception of a phenomenon by transferring understanding from familiar areas to new domains, as well as restrict our understanding of new phenomena by working as boundaries for our thoughts (Lakoff and Johnson, 1980). Cyberspace—a term coined by science fiction author William Gibson—is a widely used spatial metaphor for the Internet. We conceive of cyberspace as the place we peek into through our computer monitors when we for instance browse the web. It is irrelevant to us that all the data has to exist in the local computer's memory before it can be viewed. We think of the webpage as residing somewhere »out there.« In the same manner we think of telephone usage as direct conversations. It doesn't matter that the sound of our voices often is converted a number of times between different kinds of signals before it is rendered back to speech at the receiving end. When we talk on the phone, it is as if our minds occupy a shared space somewhere in-between here and there. Although it would imply a drift away from the original meaning of the term—also this can be thought of as cyberspace (Sterling, 1993).

Through the technological advances of graphical rendering and network communication it is now possible to dress this metaphor in colors, textures and sound. Where once there only existed an abstract idea of space, there are now concrete

(although virtual) places that we can enter and interact within via avatars. They can be general purpose meeting places like most worlds built using the Active Worlds or Onlive Traveler systems, or made for specific purposes like the Everquest gaming world. Unlike constructs in our physical world, these places have no pre-existing environment that they are placed within. They have to be thought out (or perhaps dreamed) and constructed from scratch. In the process of doing so, we encounter some rather interesting questions. How do we represent ourselves and our environment in the absence of all the benefits and limitations of an a priori physicality?

This is where the field of virtual world design emerges. There is no consensual definition of the concept »virtual world« and I have no wish to indulge in the issue of different possible or proposed definitions here. But for the sake of clarity, let me just mention that I will be talking about places in cyberspace which exist for extended periods of time and can be accessed by large numbers of participants simultaneously. They also provide some kind of spatiality including some kind of representation to indicate the positions of the present participants. There are text-based systems (e.g. MUDs) that fill these requirements and I do indeed consider them to be virtual worlds, but here I will be focusing on the graphical variety. Virtual reality (VR) will be used as a wider concept also including environments that are not persistent or distributed.

The field of virtual world design is a fairly new area of research without a strong theoretical or methodological foundation. In my personal endeavor to develop an approach to this kind of design efforts I have found valuable support in the writings of the American philosopher Michael Heim. In this text I have tried to relate Heim's theoretical framework to some of the issues that I believe to be important in understanding virtual world design in the form of four lessons. After the account of these lessons I will take a closer look at the issues of structure, function and interaction in relation to one of my own design projects, as an attempt to make sense of these lessons in a concrete design process.

INTRODUCING VIRTUAL REALISM

In 1992, Michael Heim had a lunch meeting with David Gelernter—the author of *Mirror Worlds* and *Machine Beauty*—where they discussed the growing divide between the people building computer systems and those becoming increasingly alienated by computer technology. Seven months later Professor Gelernter was severely injured by a mail bomb sent to him from the so called Unabomber, a pathological neo-luddite who targeted people he saw as advocates of the technological revolution for his sinister, sometimes lethal, attacks.

In his first two books—*Electric Language* (1987) and *The Metaphysics of Virtual Reality* (1993)—Michael Heim had written about an ontological shift that is presently occurring all around us. He talked about how we change our ways of thinking when using a word processor instead of a typewriter or how the nature of our experiences shift when we use something like VR technology. At the centre of the ontological shift that Heim was examining lies computer technology; the same technology that Professor Gelernter identified as the divider in a cultural clash that he himself became a victim of.

In his third book, Heim decided to address the problem of the growing divide. In *Virtual Realism* he develops the idea of a middle path between the utopian network idealists and the reactionary naïve realists. In the book Heim describes five aspects of the virtual middle ground. These five aspects are (1) clarifying the language of virtual reality, (2) creating a feed-back loop between the engineers of virtual reality and the public, (3) observing current shifts in telepresence, (4) cultivating premodern somatic awareness through the practice of Tai Chi and (5) developing appropriate design models for virtual worlds (Heim, 1998:1, p. 46; 1998:2).

This paper will focus on the fifth aspect of creating the virtual middle ground to see what virtual realism implies to virtual world design. It is an aspect of virtual realism that Heim has continued to explore extensively after finishing the manuscript for the book (Heim, 2000:1) and the material I have used here also includes some of his more recent writings and publications as well as notes from communications between the two of us.

LESSONS FROM VIRTUAL REALISM

Lesson 1: Technalysis

In order to know how to design a virtual world we need to understand something about what virtual worlds are and what they potentially can be. Since virtual worlds are a new phenomenon, we need methods for developing this kind of understanding. Heim has his philosophical roots firmly placed within the phenomenological tradition beginning with Husserl and then in particular following Heidegger's branch (Heim, 2000:2). Phenomenology is dedicated to describing the structures of experience as they present themselves to our consciousness. Heidegger particularly stressed that phenomenology should bring forward what is hidden in ordinary, everyday experience. In *Being and Time* he attempted to describe what he called

the structure of everydayness or being-in-the-world. To Heidegger, one is what one does in the world.

When Heim sets out to define an appropriate method for understanding our being with technology he propagates the need for an active first-person engagement as a way to achieve this understanding. If we want to investigate the meaning of a world from a human perspective, we need to focus on the interrelations to understand the components of the world, the way the nail belongs to the hammer which in turn belongs to the intentions of a user as he unmistakably heideggerishly puts it. It is in the acting in the world that its properties are revealed (Heim, 1998:1, p. 90–91).

The first-person study of new technologies also brings to words the way our reality is transformed. Heim uses the term technalysis to denote our descriptions of encounters with new technology. It is important to make these encounters explicit before the technology blends into »the invisible furniture of everyday life« (Heim, 1998:1, p. 47) in order to make critical judgments about specific technologies rather than the wholesale praise or rejections typical of the networked idealists and the naïve realists.

At the root of technalysis lies a desire to understand the values of technology based on experience.

At the root of technalysis lies a desire to understand the values of technology based on experience. Technalysis can be seen as a sort of phenomenological manuals for technological artifacts and a mirror image of an ordinary computer manual. Instead of—as in an ordinary manual—describing to a human how to use a certain technology, the aim of technalysis is to describe the changes in the mindset of a person who comes in contact with the technology. When the computer manual might say, »Whey you push button X, this happens in the computer« a technalysist might instead state, »When I push button X, this happens to me and my view of the world« (Heim, 2000:3).

I have not engaged in any low-level technalysis in the way exemplified above but I do embrace the idea of seeking understanding through experience. For me this approach led to several years of ethnographic studies of virtual worlds in order to obtain an inside view of the specific characteristics of the social interaction in these environments. At the time, much of the research presented was conducted with an outside perspective, and my results contradicted some of their findings. Studies based on media richness theory and other related theories tended to rest on a positivistic grounding and be decontextualized and task related.

While their results indicated that on-line interaction is inferior to face-to-face interaction and only suitable for simple formal tasks, I found that the interaction

in virtual worlds cannot be mapped against f-t-f interaction. By having an outside view, they mistakenly assumed that f-t-f interaction can be used as the gold standard that all other forms of interaction should be measured against. The inside view revealed that the new context makes comparisons flawed since interaction based on the unique characteristics of the medium is inherently different. I also found other assertions regarding computer-mediated communication to be too categorical and exaggerated. Both the propagators as well as the skeptics made claims about anonymity, lack of hierarchies etc. that did not hold up when compared to lived experiences. Finally, I found that mediation had been mistaken for fantasy by most people—researchers and the general public alike—who had not engaged extensively in mediated interaction themselves. They simply assumed that if two avatars interact in a virtual world, that interaction would not have any implications outside that specific context. But my experience is the reverse, while interaction mediated through avatars may be very different compared to f-t-f interaction, it's still real people with real feelings interacting. Turning off the computer does not erase any of the consequences of the interaction. If you learn something new or if you are offended, for better or for worse, you will still remember it (Jakobsson, 2001).

Lesson 2: Habituation

According to Heim, virtual worlds should strive for something between pure fantasy and photo-realistic reproduction of the physical world (Heim, 1998:1, p. 48–49; 1998:2, p. 3). Most virtual worlds available on the net today strive very hard to reproduce the physical world. The Holodeck from *Star Trek—the next generation* (a VR environment which is impossible to distinguish from the physical world) is often referenced both by developers and users as a goal to strive for. Just as I found that people with an outside view thought that interaction in virtual worlds belonged to the unreal domain of playing and fantasies, I have found that the strive for photo realism is a longing for authenticity. But reality will not be attained by technical refinement of the systems, virtual worlds become real to us through our habituation of them, just as the reality of the physical world is a result of our dwelling in it (Heim, 1998:2, p. 3).

We should also remember that computers were originally designed for organizing data, not for facilitating human communication. So there is a need to project—through our active effort—the human back into the system by focusing on the functional, rather than structural aspects of the world. The value of the virtual worlds cannot be measured by how much they look like the physical world, but by

how well they can serve the functions that we attribute to them (Heim, 1998:2, p. 3; 2001).

By doing this we open up for possibilities to transcend the physical world. By incorporating imagination and fantasy into the design process we can make virtual worlds that are more attractive by offering something extra to the participants. Heim describes this as letting us have less gravity as avatars than as primary beings. This advice can be taken literally to mean that virtual worlds should not have gravity, or only partial gravity, as the ACCD World that Heim's design team has created in the Active Worlds system. But it can also be interpreted metaphorically, staking out a direction for design of virtual worlds that allows for a touch of whimsy and playfulness, not as ends in them selves, but as means for achieving the goals of the worlds (Heim, 1998:2, p. 3).

Lesson 3: Essence

In *The Metaphysics of Virtual Reality*, Heim offers a list of concepts that are connected to the essence of virtual reality. They are simulation, interaction, artificiality, immersion, telepresence, full-body immersion, networked communications, activity/passivity, manipulation/receptivity, remote presence, and augmented reality (Heim, 1993, p. 109-128). Over time some of these concepts have been de-emphasized. The importance of full-body immersion is for instance also propagated in Virtual Realism but in a later comment on the book Heim admits that this conviction had been shattered by trying new systems with interaction possibilities smooth enough to allow the user to psychologically »slip into« the virtual world on the screen. »The photo of the author on the jacket of Virtual Realism shows me against the background of [the Active Worlds world] Atlantis. The jacket photo was the very last step in the publication process and at that time I was already groaning at the limits of my ›strict‹ definition of VR in the first chapter of the book.« (Heim, 2000:1) Part of the lesson about the essence of VR is that immersion as a psychological or phenomenological effect cannot be measured using any simple algorithm focusing on sensory input. Engagement and interactivity can give the VR sensation using standard pc equipment.

One of the primary ideas behind ACCD World was to investigate the inherent properties of the medium. (Heim, 1999, p.5) Instead of relying on concepts from other media like film, painting or theatre, Heim wanted to focus on the unique characteristics of this medium to develop its own aesthetic. He notes that, e.g., cinema is both passive and linear while vr has first-person engagement and free naviga-

Figure 1. *From the ACCD World.*



tion that is complicating the possibility for a narrative structure while opening up for new possibilities such as playability (Heim, 1999, p. 5; 1998:2, p. 5; 2001).

But although the medium is unique, it does have important similarities with other media. One of the desirable characteristics of a virtual world—playability—is a term used to describe a central and elusive characteristic of computer games. The distinction between virtual worlds and computer games is sometimes impossible to make, at least according to my definition of a virtual world, so they inevitably have many things in common. Even when a virtual world has a work-related theme, the avatars will still offer a fictional displacement. People will perceive the »lightness« of the environment that will reflect in their behavior (Heim, 2001).

The discussion about influences from other art forms brings us back to the discussion about realism. In the book *Understanding Comics*, Scott McCloud makes a very strong case for what he calls amplification through simplification. By focusing on the significant details, an artist can amplify meaning in a way that realistic art cannot. Concepts are prioritized over detail to enhance the function of the representation. Leaving realism behind also paves the way for things to happen that usually does not happen in the physical world. Just as animators know that if you are going to have a coffee cup start singing, you should concentrate on giving it the possibility of an identity rather than making it look as much as a physical coffee cup as possible (McCloud, 1999, p. 30–41).

In the same way it is possible in virtual worlds to construct buildings where doors are portals to other parts of the world, or different worlds, where walls talk to us, and where a small room can contain a much larger one. And it is probably a good idea to somehow try to prepare visitors to these possibilities through the style of the building. So while the essence of virtual worlds lies in its unique characteristics and can't be derived from comparisons with other media, there is valuable input to gain by studying everything from game design to animated film.

Lesson 4: Substance

According to Heim, western science has traditionally been preoccupied with substances on the expense of processes supporting the substances. Post-Newtonian science has, however, made a shift in this respect towards energy fields and interrelations between entities that has made new phenomena appear. One could say that it is a newfound interest in the context of substances. A similar shift can be observed in modern philosophy where the subject-object dichotomy gave way for an interest in the context in which substantial entities and psychological subjects can arise. Heidegger's notion of a world as a »context of involvements« is an example of this development (Heim, 2001:2, p. 2).

Heim argues that the same shift of focus is called for in the understanding of virtual worlds. Caught in ideas stemming from traditional architecture, we have been looking for the substance of the worlds in its building blocks. But whereas the material of physical buildings plays an essential role in the construction of physical environments, attempts to reproduce these materials in virtual environments only lead to the construction of fundamentally hollow and empty facade that imprisons the participants rather than set them free.

We have to leave the idea of designers providing a fixed environment consisting of graphical elements that we then think simply has to be filled with people to create communities. Instead we have to direct our attention to the event-based interactions to locate the »substances« of virtual worlds. »A good virtual environment, therefore, is not an object seen in and for itself but an environment that blends into the user's activities.« (Heim, 2001:2, p. 2–5)

BUILDING FUNCTIONS

Confuse

The function centered approach was taken as the starting point when I led a group of designers in the creation of a virtual environment called Confuse. By forming the name of the world from its intended function—conference use—we alluded to the fact that we tried to build a function rather than a building structure. By looking at ACCD World we could see that the reaction against structural realism into the area of structural fantasy can lead to interesting revelations but also new problems. Without a ground plane or other structural entities mimicking the physical world, especially inexperienced participants easily run the risk of feeling a bit disorientated and overwhelmed by the experience. This is perfectly acceptable and even

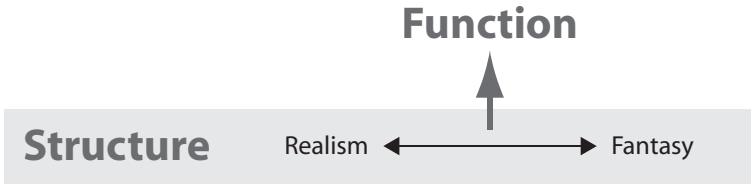


Figure 2.
The Function synthesis.

desirable in a place like ACCD World which was created to make a statement about, among other things, alternatives to structural realism.

In Confuse, on the other hand, we wanted something less confusing (excuse the pun) that still was alleviated from the clutches of structural realism. Instead of only focusing on positioning the design correctly on the line between fantasy and realism, we focused on providing functions instead of structures hoping that we had found a synthesis beyond the dichotomy of realism vs. fantasy (figure 2). This triggered us to use a standard non-descript building material that could be easily remodeled in an iterative prototyping design process. It is no coincidence that the building blocks of the world bears a striking resemblance to Lego™ since another research issue we investigated was how physical models could be utilized in the design process and the physical building material we used happened to be Lego. The function we wanted to accommodate for was meetings. We divided this into four categories, small meetings between two to four people, medium sized group meetings with four to eight people, presentations with eight to sixteen people and a general pre/post meeting mingling area as shown in figure 3.

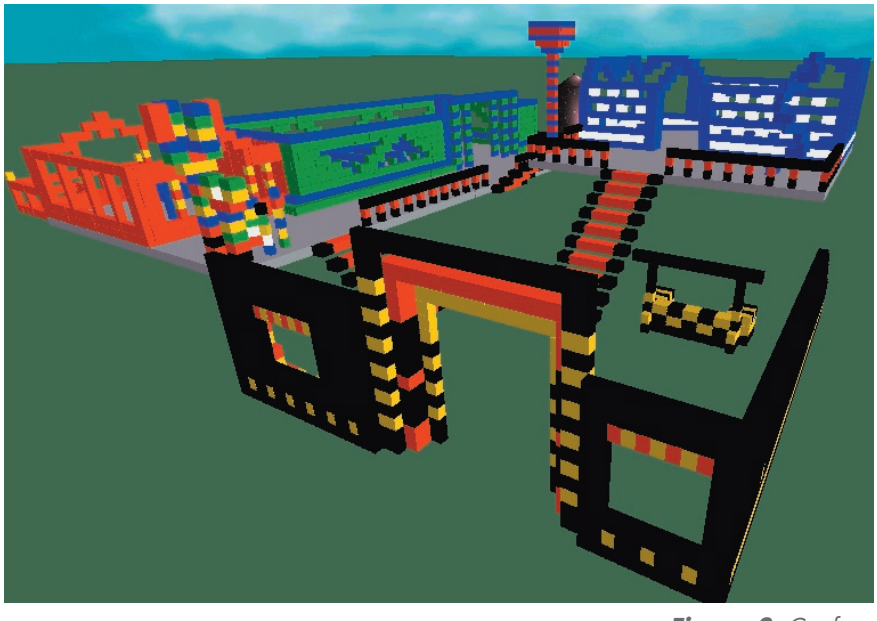


Figure 3. *Confuse.*

Figure 4. *The small room and the blue room.*

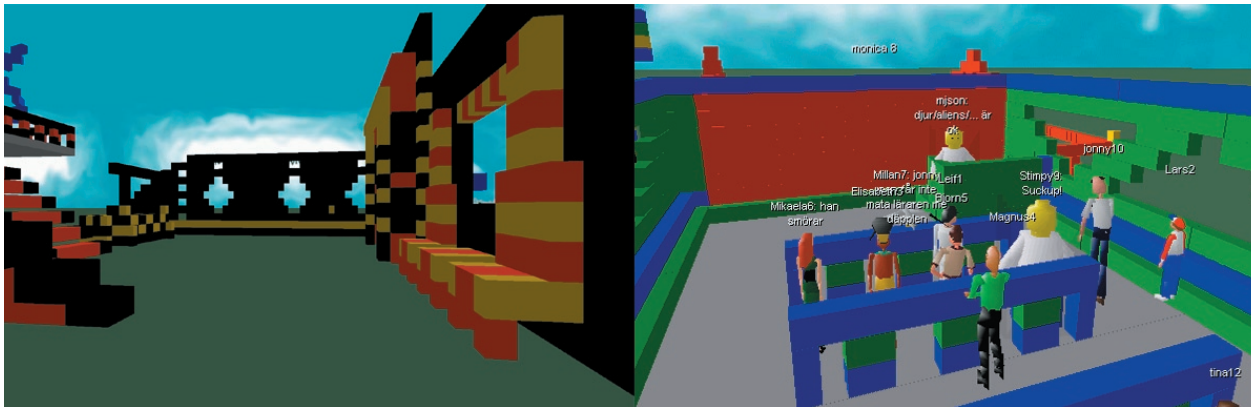


Each function corresponded to a room in our building. For the small meetings we wanted to enhance the intimacy of the meetings by making the room small. We also placed a table in the middle that we hoped would work as a subtle positioning cue to make people position themselves at an appropriate distance each other and in the field of view of the other participants. Since walls do not stop an avatar in one room from seeing what someone in another room types in the chat window, we did not make the walls solid. The holes in the walls were intended as a reminder to the participants that these walls lacked the function of enclosing the communication, that everyone in the building could »hear« you (figure 4).

At the ground floor we put a mingling area for people to interact more informally, wait for others to arrive etc. To signal to the visitors that this was something more than an entrance hall that you only pass on your way to the other rooms we put a small bar and a couple of chairs there. Although it was not possible to get a drink from the bar or sit on the chairs, we saw them as filling a purpose in signaling the function of the room (figure 5, left).

The room for eight people and above was designed as a lecture hall with one person standing on a podium in the front, facing the others. I had on earlier occasions noticed that when virtual world presentations are given in open spaces, the audience tend to move around and talk among themselves during the presentation which in turn tends to make less experienced presenters freak out. Therefore we placed railings in the room to make the audience distribute evenly in the room and stay put with their attention directed towards the speaker during the presentation. We had noticed a general feeling of restlessness in the interaction between avatars in Active Worlds and this was an attempt to bring some peace by providing a functional context (figure 5, right).

Figure 5. *The mingling area and the lecture hall..*



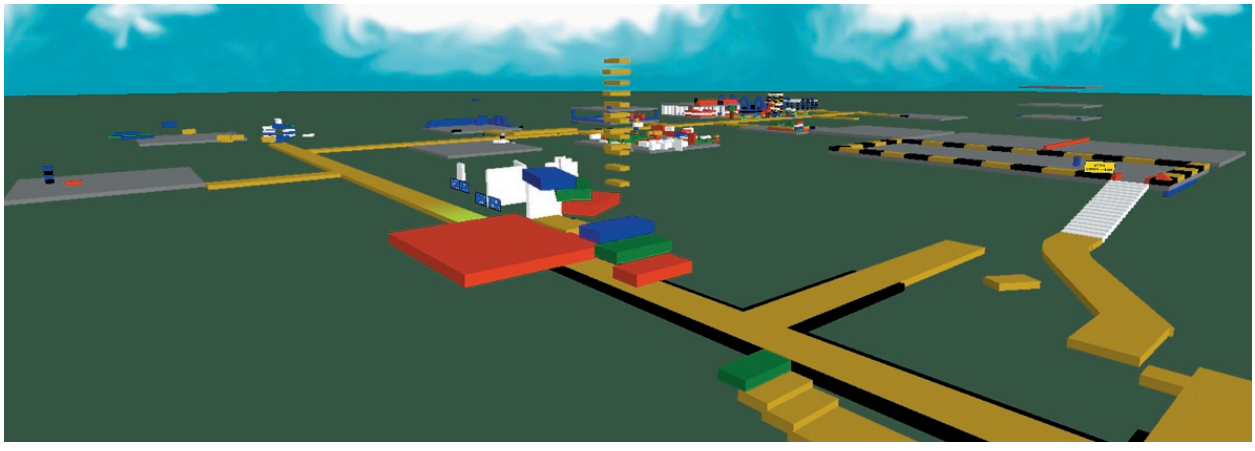
Beyond Confuse

Our idea was not to create a complete finished environment, but to create something that could be altered and added to indefinitely, but we did reach a point where we felt that the foundation was in place and we could start using our creation. That was when we realized that all was not well in Confuse. It was not that people did not like the building. Being immersed in Lego leaves very few people unaffected. But it was not utilized at all as we had expected. My first lecture there had not even started properly before I realized that there was no chance of me giving a forty-five minute presentation with the students listening quietly to what I had to say.

It slowly dawned on me that it is not possible to take a function, e.g. a lecture, out of its physical setting and implement it the same way in a virtual setting. It is not just the structural aspects that change, functions are also transformed. In fact, we had made the same mistake as the people importing structural realism but on another level. We had tried to import functional realism. So once again we find ourselves faced with the problem of having to strike a balance between realism and fantasy. At the same time as we want to make use of the participants' understanding of what it means to be in a learning situation we have to transform the function to fit in, and make use of the properties of, the medium.

Confuse never became the hit we had hoped for as a virtual meeting place. Since it only was put to sporadic use, it is hard to tell how well the specific design solutions worked in practice. The only thing I can say for sure is that the participants do not adhere to position markers in the way we had intended. I have used the small meeting room at a number of occasions, and almost never have the participants lined up around the table in the manner we had predicted. While the people I meet with often do keep »eye contact« with my avatar, they seem just as happy standing in the doorway or halfway out on the balcony during discussions. I guess the lesson here is that avatars do not take furniture as »serious« as humans do which is in

Figure 6. *View of the free building area in Ogel.*



line with the idea that the building structures are subordinate to the functions. The problem was just that we did not know how to support the function properly. The really observant reader might have noticed that while the avatars in the medium sized room (figure 4, right), which does not have any directional cues, are facing in all directions while the avatars in the lecture hall (figure 5, right) are all facing the podium. This behavior is perfectly in line with our intentions, although they did not distribute as evenly as I had planned for in the lecture room. But since the environment did not function as intended anyway, we did not use it enough to tell if this behavior was consistent.

Although the Confuse world was unsuccessful, all it needed to become a success was a reformulation of its purpose. Since we all agreed that the process of building the environment in virtual Lego had been tremendous fun, we decided to make a clone of the Confuse world, and make that clone open to the public so that anyone could have a go at the wonders of virtual Lego building. That world—named Ogel—proved a big success. We have used it for several courses in teaching virtual world construction and design, people from all over the world have taken advantage of the possibility of making their own creations in the world, and it has received honorable mentions within the Active Worlds community (figure 6).

FUNCTION VS. INTERACTION

As we learned in lesson four, Heim has suggested that it is in the event-based interactions that we locate the »substances« of virtual worlds (Heim, 2001, p. 2). This re-wording of the concept of functions opens up for a more detailed understanding of what a function actually is. One part of a function is the event and one part is the

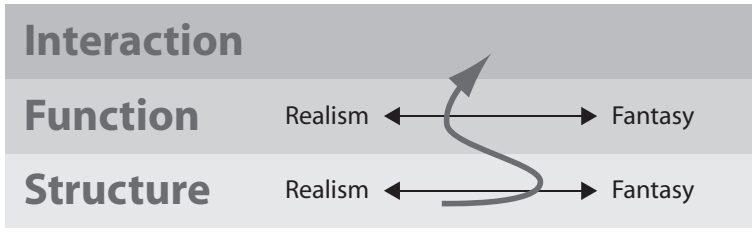


Figure 7.
The path towards interacture.

interaction. The character of the event will change from instance to instance, but the interaction is a more stable component.

All the different contexts I have studied, including conferences, design work in groups, recreational social interaction in groups and development of close social relations in and around virtual worlds share some interaction characteristics. There is a layer of interaction that is more fundamental than the layer of function. It does have the same division between realism and fantasy and the same call for balance is still valid. But the interrogation into the nature of social interaction in virtual worlds promises insights that are more general and thus can work as helpful advice in all kinds of virtual world design projects. It is time to introduce the final level of my model and the design approach that comes with it.

Interacture

The interacture approach to virtual world design entails two things. First of all it suggests that the design process should begin at the interaction level as opposed to the function or structure levels. Beginning at the correct level is not enough, however. It is also, as on the earlier levels, a question of striking the correct balance between realism and fantasy.

Beginning with the process issue, we can look at the interacture model as an inversion of the path we took in the process of coming to terms of the different levels of virtual world design (figures 7 and 8). Instead of ending up at the interaction level, this is where the design process should depart. Interaction issues are at the very core of virtual world design. Knowledge about social interaction in virtual worlds also transfers well between different contexts and affords some dearly needed stability to base the design on.

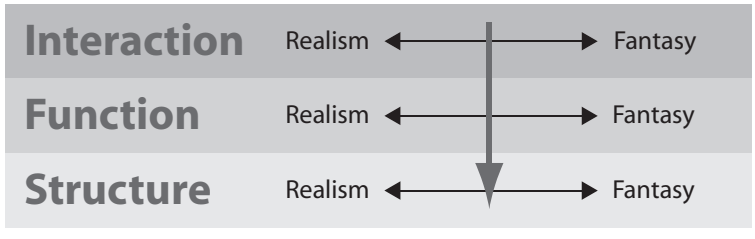


Figure 8.
The interacture model.

Then comes the function level, and as the last link in the design chain we have the issues of material structures. The interacture approach does not alleviate the designer from dealing with the issues of function and structure. The function will have to be implemented somehow, and knowing that function should be addressed ahead of structure, that the balance between realism and fantasy should be considered and that the implementation might look very different from how we »implement« functions in physical contexts has perhaps not taken the designer that much closer to a good solution, only away from a bad one.

The interaction level of my model carries the heart of virtual world design. Striking a balance between realism and fantasy on the interaction level is about understanding what is new and what is not within the social interaction. Bringing up this subject is like coming home for me since I have been spending most of my time the last couple of years doing ethnographic studies of social interaction in virtual worlds and I can talk at length on this subject. But this is not the time and place to do that, and I have done so elsewhere (Jakobsson, 2001; Jakobsson & Popdan, 1999). Instead I will conclude by noting that the interacture model is my contribution to the fifth aspect of virtual realism, the development of appropriate design models for virtual worlds. Exactly how appropriate it is, I will leave it to others to decide.

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EXPLORING THE FUTURE: ETHNOGRAPHIC FIELD MATERIAL IN DESIGN WORK

Martin Johansson

ABSTRACT

Design is about change. To make something better that has to be altered. Wanting to explore new design possibilities I have turned to ethnography to learn how artefacts are used and how activities are carried out. Ethnography is about describing things as they occur, and does not provide any direct connection to an interaction design process, if the relation is wanted it has to be created. This paper is about how that can be done.

In this paper I want to raise a voice for using ethnographic field material as building blocks for constructing possible futures in collaborative design sessions. The approach presented here shows how it is possible to bring one part of the work usually done by ethnographers into a collaborative design process. The examples given here are from different design projects, where design groups are using field material to build and tell design scenarios, as a way of exploring design concepts. My standpoint is that the ethnographic field material should be a resource during the design process and analysis should be an integrated part of that process.

The design projects referred to has been carried out at the Interactive Institute Space studio (Nilsson et al., 2000; Brandt and Grunnet, 2000; Binder and Messeter, 2001; Johansson et al., 2002). These projects have involved »end-users« and other stakeholders, participating in collaborative design processes.

INTRODUCTION

Design does not need to be informed by work practice studies, but during the last decade or two there has been a large interest for ethnography to inform design projects. The title of this paper is pointing to the need for integrating the analytic process of the ethnographer and the constructive process of the designer.

> He [the designer] does not separate thinking from doing, ratiocination his way to a decision which he must make later convert to action. Because his experimenting is a kind of action, implementation is built into his inquiry. (Schön 1983, p. 68)

In the quote above Donald Schön describes a designer's work both as constructive and as an inquiry. The tools used by the designer referred to here is an architectural drawing, a sketch, but what if we let designers build with pieces of the field material from ethnographic studies? This of course raises many questions about how to do this in practice. What are the criteria for selecting the pieces to build with? How should a design team be composed to do this? Can it generate innovative ideas?

RELATED WORK

For many years several researchers have dealt with the problem of how to make ethnography inform design, all with the strong belief that knowing about the world is useful when changing it. This paper will not discuss whether this is a valid theme, but will go into the more specific question of how a design process can be set up to integrate and learn from ethnographic work. The question of ethnography and design has been developed within the CSCW (Computer Supported Cooperative Work) community. English sociologists (Hughes et al, 1994; Bentley et al, 1992; Randall et al, 1995) and American anthropologists (Blomberg et al, 1996; Suchman et al, 1999) have been the leading voices within the community both working with ethnography informed by ethnomethodology. In Scandinavia, the majority of the researchers within the »Scandinavian collaborative design« tradition have adopted these ways of working (Kensing and Simonsen, 1997; Nilsson et al, 2000; Brandt, 2000; Binder and Messeter, 2001; Johansson and Pettersson, 2001; Karasti, 2001; Johansson et al, 2002).

At Blekinge Institute of Technology, a group of researchers and teachers have developed both a master education and a research platform following the CSCW tradition in general, and the former Work Practice and Technology group at Xerox PARC in particular (Eriksén, 1998; Tap, 2001). Together with Malmö University a

PhD education in Interaction Design has been built-up, where Work Practice and Technology is one of three possible programs. This is the program I attend, and the approach that I suggest is inspired mostly by the projects carried out by the WPT group.

In this paper, thus, I will go into dialogue with two of the English sociologists' approaches. The reason for doing this is partly because these approaches have been made very explicit, something that makes them easier to discuss, and partly because the differences in perspective might provide a richer picture of possible ways to go about.

Plowman et al (1995) points out that many researchers have stressed the importance of work practice studies in design, but that the vocabulary used is modest. Terms such as »informed by ethnography« and »relevance for design« is not saying anything about how it can be done, nothing about how design is informed. Traditionally ethnographers use »thick descriptions« to describe a practice, without suggesting ways of usage of the description they provide. Despite a large interest in the ethnography-design relation, few concrete ideas in this direction have been presented. One exception is Crabtree (1998, 2001) who suggests one approach making use of stories for design. Recently another direction has won the interest of several researchers in the field, namely the patterns approach first introduced within architecture by Christopher Alexander. This approach is represented here by Martin et al (2001) and takes on the task of creating patterns in the sense of »families of resemblances« out of concrete instances from work practice studies.

ETHNOMETHODOLOGY

The ethnography that has dominated the CSCW community is informed by ethnomethodology (Shapiro, 1994). The English sociologists offer good arguments for ethnography to inform design, and have been stressing the importance of acknowledging social aspects and the complexity of work as important input for technology development. They have continued on the ethnomethodology track laid out by Garfinkel, and state that they do not make any assumptions but only report versions of what has been studied. Ethnomethodologists claim that everything is out there, and the only thing they can do, is to give descriptions of existing practices, offering »accounts« of »the actual ways in which work is done« (Blythin et al, 1997). They do not consider themselves as designers, and do not want to give advice on design, but they can tell stories about what they have seen.

From detailed studies done by the English sociologists we have learned a lot about how technology actually is used in praxis, and this knowledge can in my view

be of value when developing new technology. Hughes et al (1994) describes this kind of usage of ethnography as »re-examination of previous studies« in contrast to different ways of doing studies for the specific design case. We can learn from it, but we do not come any closer to the context we are designing for. Button and Dourish describes the CSCW ethnography as a critic of technology and wants to see »how design can productively learn from ethnomethodology.« For the purpose of finding these new ways for ethnomethodology to inform design, they make a theoretical exercise in dividing possible ways of establishing a relationship between ethnomethodology and design (Button and Dourish, 1996). They found three categories: *Learning from the ethnomethodologist*, *Learning from ethnomethodological accounts*, and *Learning from ethnomethodology.* It is the third one I will elaborate on here. »This alternative view is less concentrated with what the abstraction (or representation) is, in itself, focusing instead on what it can do and how it can be made to work.« (ibid) In a later paper Dourish and Button (1998) use the phrase »analytic mentality« to describe the differences between ethnomethodologists using ethnographic techniques, and others using the techniques. They exemplify these differences with the manner in which »selection of phenomena« is done and how »topics for investigation« are chosen (ibid, 1998). From my perspective it is this mentality that is the burning point. It is hard to use detailed accounts in design, since they tend to lead us into searching for problems that can be solved, but for a team of designers to apply the analytic mentality of an ethnomethodologist seems to be something that in a fruitful way can be weaved into the design exploration.

There are few contributions that report a bit more practical ways of making use of the ethnography/ethnomethodology to inform design. Crabtree (1998, 2001) uses story telling as a »common sense method of linking ethnography and design« (p. 236). Crabtree wants to give ethnography »a constructive role« in design, however still not regarding the ethnographer as one of the designers but leave the design work to designers (p. 208). The design setting Crabtree sees is a collaborative setting where persons with different competencies meet and each taking on the parts of the design process that they are trained to deal with. In Crabtree's approach, the ethnographer's role is to tell stories to the design teams, stories that together with the designers can be transformed into scenarios useful in the design process (p. 224). The ethnographer takes on the role of the »bricoleur« bringing the stories he has into the design discussion, when appropriate (p. 210). To relate this view the categories of Button and Dourish (1996), I would regard Crabtree's story telling as Learning from the ethnomethodologist: In this approach the ethnomethodologist is actively taking part in the design sessions and can choose what stories to tell. He does not deliver a package of accounts, instead he uses his understanding and over-

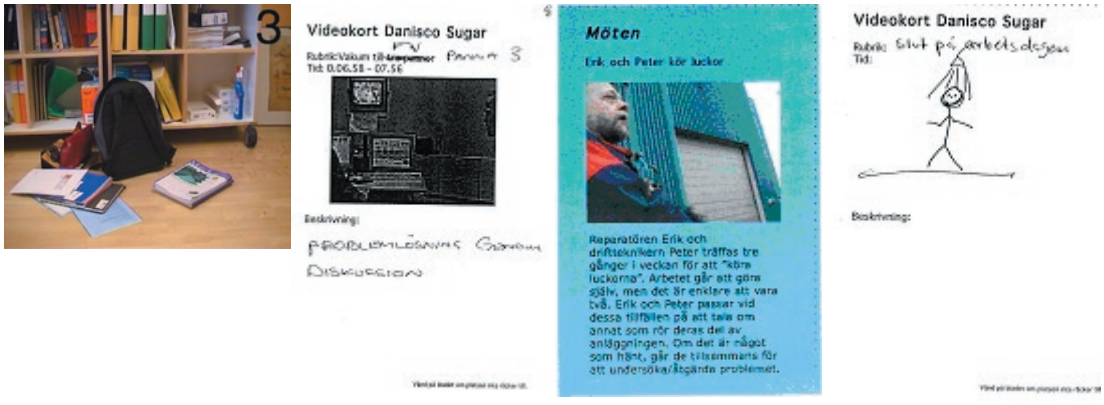
view to choose when and which story to tell. Since Crabtree keeps a strict division of labour between ethnographer and designer, the designers do not get to navigate the material themselves but have to rely on the ethnographer. The approach offers a way of using the field material actively and the design process is given a prominent position, to which the field study can contribute when suitable.

Dave Martin and colleagues (Martin et al., 2001) have chosen a rather different approach, using pattern-thinking building on the work Christopher Alexander did within architecture. Lately many computer system developers have adopted the patterns thinking as an extension of object-orientation. To continue to use the framework of Button and Dourish this approach would classify as Learning from ethnomethodological accounts. »The idea is to first find recurrent phenomena and to make these patterns available to designers« (Martin et al, 2001) The patterns are to be drawn out of accounts, and a collection of patterns could be more easily accessible to designers than the accounts themselves. Martin et al (2001) presents two examples of patterns, »artifacts as an audit trail« and »multiple representations of information.« The latter pattern consists of accounts from two work place studies, »ambulance control« and »air traffic control.« The accounts within a pattern are not limited to having only one author; instead the pattern is better the more accounts and the more authors there are. Each pattern aims to bring light upon how different artefacts are used in the specific settings (Within the »multiple representations of information« pattern one account included »The incident stack, the dispatch selection screen, the vehicle availability screen and the automatic vehicle location screen«, the other included »flight strips and radar screens«). Whilst Crabtree is taking an active role in the design process, Martin and colleagues highlight experiences about artefacts from earlier studies through the patterns. It is up to the designers to see possible similarities to what they develop, but when they do, there is a set of ethnographical examples showing the benefits and the drawbacks of artefacts used in different contexts. Martin and colleagues have developed a structure for the presentation of patterns that aims to make findings from studies more easily accessible for designers, and encourages others to use this structure to describe their findings.

BUILDING BLOCKS

The Building blocks referred to here are video cards (eg. paper cards that has associated video snippets to each card). At some instances we have used text to describe an occasion illustrated with the video card, at other times we have only used references connecting cards with images to video clips.

Video Cards from different design sessions.



The design process that we work within most often starts with some kind of request from an external party, and it typically involves work place studies and ideas about technology from the external party. This approach has in different manifestations been used in many projects during the last few years. What binds it together is that we use sets of tangible paper/plastic cards representing video snippets from the field material. The idea with the cards is that they are used to build stories about possible futures. Connected to each card is a video snippet showing an activity and providing a context. The number of cards in the sets has varied between 20–40 cards, and the video snippets are between 40 seconds and two minutes. Our work with video cards is inspired by the »Video Card Games« developed and described by Buur and Søndergaard (2000). The video material that Buur and Søndergaard use in their video card game originates from a usability test workshop, during which the users were asked to install prototypes of heating regulators. Selected video sequences from these tests were viewed and »made sense of [...] for usability matters« (p. 63). We adjusted the cards to try this technique with ethnographic field material in a design concept exploration workshop.

The idea is that the participants can use fragments from the studies to build their scenarios, and while building look into the pieces in more detail. We let the workshop participants work with the design material as they find it suitable from their competent view. But it is in our case not merely to »play around with the truth« (as for example Gaver et al., 1999, suggests for their cultural probes), but rather to use »true« images of existing practices as »building blocks« for creating visions of the future. In our work the field material is not treated as »data« but as something to collaboratively explore and work with to build visions about the future (Johansson et al, 2002).

Most of the stories have been possible to create with the cards provided, but it is also necessary to open up the possibility for adding things that are not found in the

Pictures from workshops where the cards have been used.



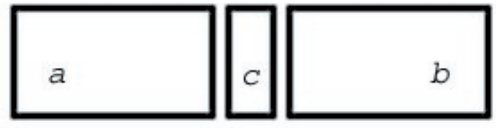
design material. We have worked in different group constellations: sometimes with many stakeholders, and at some occasions only with the persons that we have followed, the future users. In one of the pictures above, a process operator participating in a workshop has made a drawing and added an activity that the group found important for their story.

PRACTICAL WAYS OF BRINGING ETHNOGRAPHY INTO DESIGN

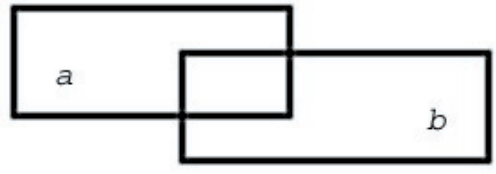
The three approaches presented in this paper all have in common that they strive to let design be informed by ethnography. Both the story telling and the pattern approach have in common that they bridge a gap between ethnography and design. The »building block approach« differs in that it does regard the relation between ethnography and design as gapless. We have taken one part of the analytic process usually done by ethnographers and brought it into the design process, making it a collaborative task. To once again use the categories formulated by Button and Dourish, this approach would fit the Learning from ethnomethodology, which is different from the other approaches referred to.

In the Learning from the ethnomethodologist category exemplified with Crabtree, the ethnographer tells a story from the field when he finds it appropriate in the design process. The ethnographer becomes a mediating filter and the only access to the field. In the Learning from ethnomethodological accounts category, Dave Martin suggests the pattern approach, which means that what is available during the design process is patterns illustrating concepts that the ethnographer has selected. In the figure on the following page, C stands for an interface between ethnography and design. What both Crabtree and Martin do is to hand over wrapped packages that they think represent what the design process needs. The Learn from ethnomethodology approach, as it is presented in this paper, sidesteps the C in the figure. Instead of making extensive analysis of the material, and bringing that analysis into

Learning from the ethnomethodologist, Learning from ethnometh. accounts:
The ethnography process (a) is translated by the ethnographer into a format (c) that can be handed over to the design process (b).



Learning from ethnomethodology:
The ethnographer and designer roles meet in each others' processes. Ethnography (a) is not finished when design (b) starts. Ethnographic analysis and design exploration are interweaved.



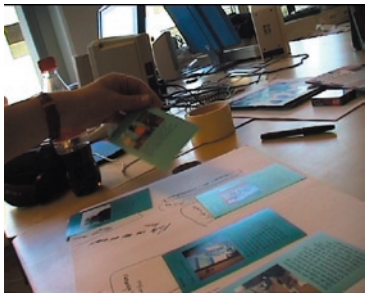
the design process, this approach suggests that it is the process of analysis that is brought into the design process.

ANALYSIS IN THE DESIGN SESSION

The building block approach is meant for a Collaborative Design setting and suggests an explorative design process where ethnography plays a central part. We are using the field material for »maintaining reference to the context« (Buur et al, 2000). The intention is to create a design lab where the participants trained in ethnography and the participants trained in design collaboratively can go about exploring field material in an integrated process of exploring design possibilities.

The following example illustrates how the analysis can be done during a design session. Before this design session an ethnographic study has been carried out. A workshop with process operators from three different plants has been held, and during that workshop some interesting themes have been identified. One of the themes was meetings, and raised the question about how meetings take place, how they are established. Examples of other themes were »Rhythm« referring to differences in tempo during a process cycle and »Machines and things« raising the question of how a machine can communicate with the process workers. For this design session we have invited designers who are familiar with process industry to look closer at how people meet in the process plant studied. Some of the ethnographic field material has been selected and made available as video snippets (approximately one minute long) and video cards labelled »Meeting cards.«

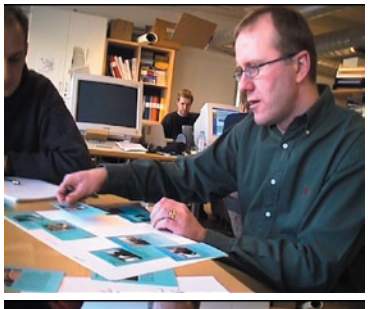
The group starts by dividing the »meeting cards« between each other, and agree on trying to make up a few categories, that can be the basis for some scenarios. After sitting quiet for a little while going through the cards individually, a discussion starts.



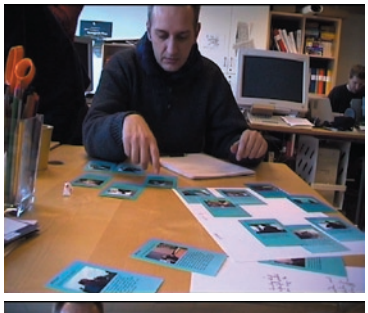
A. *Designer 1: I can start. I have seen something about predictable meetings and unpredictable meetings.*



B. *Designer 2: So there are places where you can always meet… Perhaps, since [designer 3] has sorted them… Here we can see [a user] engaged in two equally important activities, and here someone finishing an activity. Something about having a place in different circles [group constellations typically forming around different problems] and also be a peripheral part of a circle.*



C. *Designer 4: This is about the physical and the mediating.*



D. *Designer 3: This is something planned and something unplanned.*
Designer 4: This one is connected to that one [pointing at two cards, looking at Designer 1 who nods].



E. *Designer 2: There are a few things growing out of what we see here. It is something concerning how a meeting starts and how it is ended.*

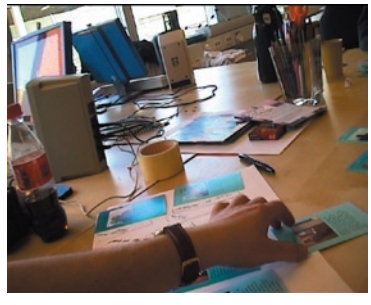
The discussion continues, and after approximately two hours the group has started to build a more coherent scenario.



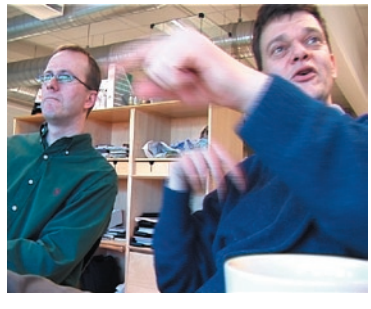
F. *Designer 2: You could imagine an intercom, where you meet on channel 27.*



G. *Designer 3: That's very much a chatroom. […] It has to be the circles that stand out, that you can control.*



H. *Designer 1: It should not say the names of the people but of the circles.*



I. *Designer 2: Do you always speak to everyone in the circle? I don't think you do.*



J. *Designer 4: I think it is just like when we are sitting here now. You don't speak to one person at the time, you just speak.*



K. *Designer 3: It is also an obligation. If you don't talk about what's relevant, you get lynched. […] And I think it is unwise to talk only about speaking as the only way of communication.*

In this part of the session the field material is not worked with directly, still it is a continuation of »the circle«—a concept drawn from the material—and when imagining an intercom system, it implies a direct reference to the context. The way of using the term »circle« rather than for example »work groups« means that the design group has to establish meaning and that they are still working with the exploration rather than defining the phenomenon. Still the naming helps the group stay with the phenomenon and to focus on it. This can be compared with what Blumer (1954) calls »sensitizing concepts« (p. 7) that gives »a sense« and »guidance« in creating ethnographic categories or instances.

In I and J the designers express what they »think.« Designer 2 imagines a possible future system and verbalizes a hunch about how the praxis would look if this system were introduced as currently described. He starts by formulating it as a question, that he gives a possible answer to himself. Designer 4 gives his possible answer that contradicts what designer 3 suggests. The reference he makes is not to the field material, but to the situation he shares with his colleagues. Designer 3 continues on what designer 4 started, with strong words explaining what he would expect from the process workers if the system were used in an inappropriate way.

The very next thing designer 3 says is back in the technology discussion; to limit the discussion to spoken communication is not a decision that should be made at this time in the design process. This is an important point, since a designer has to keep in mind that the ideas he presents should be possible to do with technology. This constraint can be dealt with by thinking in examples of how things could be

done. The comment by designer 3 helps the group remember this, and everybody agrees that this is not yet decided.

The shift between what can be seen as an ethnographic analysis and a design exploration happens seamlessly. To make distinct categorizations would mean that the dialog has to be broken down sentence by sentence, or even shorter fragments. On the other hand, a meaningful summary of what was learned about the work practice of process workers would be possible to do. Likewise it would be possible to identify design leaps, and how the design team works with the concept ideas.

VIDEO CARDS AND SCENARIOS

We have used our video cards in different ways on different workshops. On some occasions the groups lay out the scenarios as storyboards, others use the cards for indexing what they base their scenario on. In the following example the latter is the case. Three persons are trying to look into R's life, and more specifically how he goes from »working mode« to »private mode.« We enter the workshop when the group is discussing R is in his music room, playing his guitar. Previously the group has made categories, and this card is put in a group of cards where R is by himself in his leisure time.



Designer 5: We need to try and find a situation where we can come up with a thing that he can use.



Designer 6: I find this situation rather interesting. It differs significantly from the other situations where we see R. There [in the other situations], R is very much a businessman.



Designer 5: What can we call this situation?

Designer 7: They have music in common.



Designer 6: The private life of R. No… The music in R's life… The music as a sanctuary.

Designer 5: The music as a sanctuary.

Designer 7: The music as a sanctuary. Perfect.

To start building on a scenario, the group begins with identifying one of the cards as interesting, because it differs from the other cards where R is dressed as a »businessman.« They try to find a good label for the situation where the scenario will start. Designer 6 thinks out loud »The private life of R. No. The music in R's life. The music as a sanctuary.« The other two designers both immediately pick up on the »music as a sanctuary« suggestion. This is a typical naming process. In this way the group tries to find a working way of talking about their scenario. In the following short extract of a discussion, that takes place about fifteen minutes later, we can see that Designer 6 who suggested the »music as a sanctuary« continues on what Designer 7 starts making up (without any direct reference to the material). Designer 6 reconnects what has been said to the video cards.



Designer 7: What has happened before? What does the prologue to this look like? Does he do this every day? I don't think he does it on Fridays or Saturdays. Then he eats with his wife, otherwise she would be really angry. Or is it when it has been one of those really bad days, »Now I have to hit the guitar.«



Designer 6: I think it is like you say, the last thing. He describes a combination of beauty and aggressiveness. This is the opportunity that he has.



Designer 5: Should we write this down?

Designer 7: I can be the secretary.

Designer 6: I would like to use this card, where he describes… And this card, too. They are connected.

The Designers use their impressions and own experiences to fill the »gaps« that appear in the scenario. This can be seen as a process of familiarizing the scenario. They are trying to make the scenario believable, or put in other words, to become comfortable with the scenario. My guess would be that it is easier to see alternative scenarios once the initial is internalized. So far the scenario has not involved any new technology, and what the group will find out later is that they don't want to ruin this scenario with technology, so instead they transfer it to a situation when R is on business trip, trying to keep some of the qualities that was found in the exploration of »the music as a sanctuary.« Once again we can follow the shift between using the cards for driving the scenario and »free interpretations« made to make the scenario useful for design purposes. As these short design session extracts show the groups are staying well grounded in the practice, along the way of envisioning possible futures.

ETHNOGRAPHY IN DESIGN

Ethnomethodology strives to tell their stories from a »native's perspective«, explaining how work is socially organised from the perspective of itself; in its own terms. This I find most attractive for the design process. Furthermore, early ethnographers (e.g. Robert Park in 1920 cited in Crabtree, 2001, p. 188) advocated »first hand observation.« In the approach presented here, the design team (including ethnographers) can get »first and a half« hand observation from the video snippets

(a half, due to the fact that it is filtered through a video recording and due to the selection out of the entire field material). And in this approach the ethnographer's role would be, rather than being the analytic of the team, to take on a role of keeping the discussion grounded in the material and make sure that the material is thought about from the perspective of »the natives« (others can of course also do this, but experience from previous work practice analysis seems to be a good qualification). This proposes a design process where all the participants take on the roles as designers. The involvement of ethnographers in this process is of crucial value due to their analytic experience. Or as put by Shapiro (1994): »But they [the designers] have no magic means available to them for reconciling orthogonal perspectives or working through the detailed consequences of social scientific studies. It seems odd to impose the entire responsibility for the redesign of the work on systems designers while those whose specialty is supposed to be the analysis of work run for cover.« Technology developers often have strong theories about »the world« and how things should be done, and these theories are often motivated by efficiency. The sociological pragmatic perspective that the ethnomethodologists have brought to sociology is in my opinion one of the things that we could learn from the ethnomethodologists and apply in the way we work with technology development. The role of the design artifacts (e.g. video cards) is that they function as the building blocks, and due to what these building blocks are loaded with, the design team will be confronted by the work practice, all through the design process.

Ethnomethodologists have successfully developed ways of working with field material, and here we try to bring some of these ways into the design process. The ambition is to let the work in the design process stay close to the field material, striving to adopt an ethnomethodological way of looking at work. The notion of »seen but unnoticed« is sometimes used within ethnomethodology to explain the benefits of careful observational studies, but the same words could be applied to design. Talking about mockups and prototypes, Ehn reflects over what makes these design artifacts good: »If the design artifacts are good, it is because they help users and designers to see new aspects of an already well-known practice, not because they convey a theoretical interpretation.« (Ehn, 1988, p. 113) As a designer one looks at the existing practice for the purpose of change, and what one wants to see is the »seeds of the future«, well-known things and practices that are just waiting to be noticed. The approach presented here is practical, perhaps pragmatic, explorative by nature, and picks fragments of ways of working from one discipline and transfers it into another. Button and Dourish explain that »learning from ethnomethodology« is »not a link between theory and practice, it is a link between theory and theory« (1996). I have claimed that the approach presented here is of this third category, but comple-

mentary to linking theory to theory, practice has been linked to practice, and design has »learned from ethnomethodology.«

DISCUSSION

In this paper I have advocated the active use of field material in collaborative design workshops. The material that has been used in this case is video. We could have chosen other means of bringing the field study into the design session, but we have not since video has suited our purposes well. Video as a material is very rich, a short snippet can contain several different fragments for a story, it can provide a picture of the context, and mediate feelings, it can show one or many activities, and so on. It is not the aim of this paper to go into detail about how to select video snippets that can become design material, but it might be fruitful to describe some of the thoughts about the material. The basic principle that we have worked with is that the work practice should be highly present in the design work. The context should be obvious, and activities taking place made observable. Often we have invited the persons studied to participate in the design sessions, and they have been able to add things (as in one of the pictures above) to what is represented in the material. We do not regard this as something we failed to include in the design material, but as confirming that the design material works and at the same time opens a possible way for these people to make their voice heard in a rather concrete way. Ehn explains that design artefacts bring »[…] earlier experiences to our mind and it ›bends‹ our way of thinking of the past and the future.« (1988, p. 110), and this is exactly what we strive to achieve. In a design session the nature of the design material and the task at hand opens gateways to learning more about the work practice. Using video snippets as building blocks in the exploration of future possibilities keeps the process in a firm grip by the praxis studied.

A question that might arise from the use of this kind of approach, is whether the field study could or should be altered to match this way of working? Currently we are working with this approach, but with field material that has been gathered for purposes other than our current design cases. This means that the approach gets more similarities with Martin and colleagues' (2001) »re-examination« of previous studies. This slightly changed approach might be better suitable for large user groups, where the field material can function more like personas, than the users.

Here I have used the categorization borrowed from Button and Dourish (1996) as a theoretical tool to look at differences between different approaches all striving for ethnography to inform design. In practice the categorizations are not isolated ways, most approaches have traces from all three categorizes. Still I think that the

categorization can help in seeing differences and similarities between different approaches to work practice based design, even though I in this paper have chosen to present only one example in each category. From my perspective, the reason for design to be informed by work practice in the first place was the »grounding in praxis« and what we can learn from ethnography informed by ethnomethodology is primarily the »analytic mentality«, breaking normative thoughts and preconceptions about how the world »should« be.

In this paper I have stressed the active use of ethnographic field material in the design process, in order to combine the analysis and the design into an integrated process.

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THE WAY THINGS GO: SOME PERSPECTIVES ON DESIGN OF FUNCTIONALITY

Per Linde

Der Lauf der Dinge, or The way things go, by Swiss artists Peter Fischli and David Weiss is a kinetic installation/film shown at the Documenta exhibition in Kassel 2001. The piece is a recording of a cause and effect chain reaction, in which a 100 feet long series of physical interactions between everyday objects like tires, ladders, shoes, bottles and so on folds out like domino bricks making each others fall. For thirty painstaking minutes one object after the other falls down by use of fire, water, and chemical reactions in fluid liquids or sheer gravitational forces. The film is just as thrilling as a Hitchcock movie with the difference that it's objects that are the actors instead of people. They animate the universe of objects by completely removing them from their contexts of use in our daily lives and then restructure their relation to one another. I don't think the idea is to neither glorify nor alienate artefacts, but instead the film makes some new perspectives on the functionality of everyday things by simulating their self-autonomy.

The art world is full of examples where the appearance of objects is put in a phenomenological space far beyond their original context of use. This is quite a contrast to a traditional design perspective, in which form follows function and the main aim of interactivity is user-friendliness. For sure we want products and digital systems to function in a coherent way, let's take that for granted. But user-centred

design might no longer only be about affordance, ease of use and transparent interfaces. The idea of transparent interfaces could be seen as an attempt to minimize the gap between user and machine.

English interaction designer Anthony Dunne (1999) takes on a different approach in trying to illuminate that very gap. His idea of hertzian space refers to the notion of electro-magnetic radiation leaking from electronic objects. This leakage is invisible to human sight, but could be made visible thus stressing the problematic conceptual level of materiality/immateriality. The nowadays less used concept of cyberspace could be seen as an attempt to spatialise activity in distributed computer networks whereas the notion of hertzian space is real and actual. Dunne (1999, p. 30) cites Paul Virilio in reflecting the human factors tradition; »interactive user-friendliness … is just a metaphor for the subtle enslavement of human beings into ›intelligent‹ machines; a programmed symbiosis of man and computer in which assistance and the much trumpeted ›dialogue‹ between the man and the machine scarcely conceals the premises; the total unavowed disqualification of the human in favour of the definitive instrumental conditioning of the individual.«

By mimicking the already set assumptions about reality, the design object in the human factors tradition becomes a matter of understanding rather than interpreting.

In writing this I can feel the probability of product designers, engineers and industrial designers leaping back in disgust when confronted with the quotation.

I, too, see Virilio's outburst as an extreme. Of course we have to cope with usability somehow, because it's in some sense all we got and I don't find it constructive to drag the usability concept into a dialectical battlefield. I also find the HCI tradition full of rich methods and a sympathetic concern for users. During the CSCL 2001 Yrjö Engeström, specialist in activity theory, while analysing the health care system shouted out »For Christ sake's there's a patient in the bed«, and that's for sure. But let me continue for a little while, because I really think the idea of user-friendliness has some trade-offs.

In search for transparency in interfaces you also get a naturalization of object/system and their embodied meaning, and they do embody meaning and ideology. That meaning is not inherited by natural materials, but man-made and artificial.

Rick Robinson (ref?), in a review of Donald Norman's *Things that makes us smart*, states that; »…user-centeredness is not just figuring out how people map things, it absolutely requires recognition that the artefacts people interact with have enormous impact on how we think. Affordances, to use Norman's term, are individually, socially and cultural dynamic.«

Dunne goes on to analyse the concept of innovation, though not using the word explicitly. To achieve transparency most of the time presupposes relying on pre-existing cultural patterns. The computer's desktop metaphor is a typical example of that. Steven Johnson (1997) claims that »radical break-throughs are anomalies in fossile cultural archives.« Maybe that's so, but they tend to stay around for quite a long time. By mimicking the already set assumptions about reality, the design object in the human factors tradition becomes a matter of understanding rather than interpreting. This makes our confrontation, or use depending on perspective, of machines easy and adaptable. But it also can hinder the huge potential of new technologies to provide new aesthetic meaning in everyday life. We need recognition, but we also need counteracts if design is not to be seen just as an incremental mass of new products and systems, because that mass is already approaching critical density.

By contrasting ideas of recognition to those of interpretation and perception we can also somewhat soften Virilio's harsh remarks about enslavement. The notion of cognitive clarity in interfaces surely has perspectives of conditioning the user. While using familiar settings for recognition there is a risk of also reproducing the embodied meaning of artefacts. Dunne refers to this as a mimetic approach, which very well can lead to behavioural circuits.

While some of the significant properties of digital media is ease of stretching, configuring and juxtaposing, the human factors tradition often reduces openness of systems and media for the sake of user friendliness and ease of use. He makes up a couple of provoking pairs of distinctions to illustrate this and at the same time propose some alternatives. While relying on pre-existing patterns for transparency, technology very much becomes like domestic pets conserving our views of reality instead of widening perspectives and augmenting possibilities for change. Dunne puts this in contrast to a »technology-as-alien« perspective. This means that he quite boldly puts forth a user-friendliness/user-unfriendliness distinction pair. I don't see that suitable for literal interpretation. The meaning is a shift of focus from the idea of transparent interfaces to providing aesthetic experience and/or critical reflection through the use of electronic objects. The means for this is a form of gentle provocation.

DESIGN AND TEXT

Very interesting for me is Dunne's analogy to literature, where the poetry has that role of constructive user unfriendliness in relation to big brother prose and even more to everyday language. While everyday language has an instrumental function

most of the time with a great amount of transparency, poetry is at the other end of that line. But confrontations with poetry can make the reader more aware of language itself. Also the amount of ambiguity calls for a more active identification of the self in relation to the world and other people. The strange and yet comprehensible voice speaking in poems are not what we would expect from the paper clip guy in Words, but maybe the idea is not so bad. Utilitarian perspectives must be at forehand in a lot of applications for sure. But there are many cases when designing interactions could benefit from practices of art. Especially when it comes to building new spaces outside of the screen we'll confront new environments, where a sense of wonder would mean as much to users as immediate comprehension. Designing for participation is also a way of caring for users.

Danish poet Pia Tafdrup (1997, p. 107, my own translation from Swedish) writes; »If the poem could be something else, then it wouldn't matter, but the poem is what nothing else is. Therefore it's about time to demand that the readers invest something of themselves. Art can't be created by meeting conventions or by the expectations of the audience.« This is not arrogance; it's a call for participation. And technology has potential for producing new kinds of spaces, which will be »what nothing else is.«

This self-investment of the user/reader is also according to Espen Aarseth (1997) part of what distinguishes what he calls cybertext (for instance hypertext, adventure games…) from traditional literary text. While most reading requires interpretation, Aarseth claims that »during the cybertextual process, the user will have effectuated a semiotic sequence, and this selective movement is a work of physical construction that the various concepts of ›reading‹ do not account for.« He even coins the term ergodic literature, derived from the Greek word ergon, meaning work. So, if a non-trivial effort is an essential property for the act of »reading digital literature« compared to traditional text, it could mean that the non-transparency is equally essential. Personally I think that part of the reason why literature has made a less impressive transference of text to digital media than, say, art, music or theatrical events, is that many projects has mapped the act of reading in a too direct way, to a medium that is completely different.

Dunne remarks as how this kind of »gently provoking design« sometimes can be defined as »text« according to Roland Barthes definition: »a space of chains and layers of meaning between the object and the viewer, continuously expanding with no fixed origin or closure« (Dunne, 1999, p. 39).

Barthes is known for his »death of the author« metaphor, a reorganizing of the linguistic order where the imaginative efforts of reader and writer are equally high. In a design context this could be analogue to a view where meaning of objects arises

through user interaction, not in a pre-conditioned mapping. According to Dunne this is applicable to electronic objects very easily, because their components can bee freely arranged. The plasticity and the re-configurable form is also one of the essential properties of digital media in general.

The concept of interactivity is closely linked to that of experiencing functionality. I'd like to recall Jean Piaget's definition of play when it comes to pinpointing experience in relation to mimicking the already known or self-explanatory. He understood play as an adaptive action in contrast to imitation. Being situated or adapting to situations is to be seen as a combination of imitating and playing. While imitating is assimilation of presumptions of the world outside of ourselves, play can be seen as a letting go of the world and re-arranging it in accordance to our own preferences. Interactivity can increase by play in a self-reflexive manner. That space for reflection is somewhat underestimated in the debate on interactivity. Whereas a lot of effort is put into making interfaces that have a frequent and sometimes intense interactivity, the notion of stopping time in order for expansion of the users' own time is often mentioned as an interruption of task and not welcome. Very often technology calls for repetition where art calls for wonderment of the unique; a uniqueness that pre-supposes participation and reflection. Digital media is well suited for this kind of self-creation and it should be an integral part of usability perspectives. Not being a specialist on Aristotle I think that his poetics is a hard orchestration of the narrative and maybe not an obvious role model for interactive narratives. Greek drama before him had a more episodic form and that holds true for many orally oriented cultures as well. In some sense it's the modern time starting with Aristotle, time now already passed.

THE WAY ACTANTS GO

Part of the amazement for Fischli's and Weiss's art piece, mentioned in the beginning, is of course an underlying expectation of what rubber tires do (they make the friction of the wheels to the ground manageable while moving) and what they are (a hybrid of matter). This expectation is grounded in a use of language where reification of artefacts is natural. Objects are monadic entities clearly separated from subjects. Their »thingness« is what makes alienation between an experiencing subject and their inherent immateriality a goal for soothing recognition and constraints in perception.

Instead of closing the gap between user and machine it might be fruitful to take a look at the very nature of that gap from still another view, and also at the tightly coupled dichotomy of language and things.

Our spontaneous use of language depicts things in an image like manner. Words seek for correspondence to their sharply disjointed points of reference in nature. They are divided by a gap left for science, not to make graspable, but to reduce by a reliable use of methods.

French sociologist Bruno Latour widens the perspective of artefacts as objects in a convincing manner. For an example, see *Pandora's Hope* (1999, chp. 2) where he describes the correspondence of words and things as confusion. He means that such a presupposition focuses on the outer extremes (language/things) of a chain with many links. The term reference is what he uses to coin matter (nature), which gradually moves along the chain to form (that is language, in Latour's terminology form might best be interpreted as a knowledge representation, not a spatial arrangement of shape). References circulate along a series of representational transformations by use of scientific methods and instruments. According to Latour those references circulate along the whole chain. So there is no singular meeting point, in the form of a distinct phenomenon such as the Kantian philosophy makes us believe. The different transformations bear no resemblance to each other but the coherency of the different stages, of what we call things, depends on how well the steps are articulated.

The quest for correspondence is quite different in this perspective and by acknowledging the transformations we can see, not only natural matter, but even more the designed object in another light. Along the path of circulating references new conditions emerge and even actual and measurable spaces, such as in the case of Dunne's herzian space. The electronic object is not just a reified tool, but also an activator of both conceptual and real environmental levels.

Closing in on perception in contrast to recognition was also a concern for Dewey and the American pragmatist movement in their attempt to promote active learning as opposed to instructional knowledge representations. Pragmatism is self-organization as opposed to authorative truths and I don't think interaction designers believe in objective truths when it comes to situations of use. This is a sort of backwards interactivity interacting with the self instead of the given and it's a part of creativity that should be supported not instructed.

Dunne (1999, p. 60) quotes Jean Baudrillard when exemplifying such extensions of use: »Consider the way a camera is used now. It's possibilities are no longer those of a subject who ›reflects‹ the world according to his personal vision; rather, they are the possibilities of the lens, as exploited by the object.«

So, Dunne's concept is by no means hostile to the emergence of electro-technical environments, nor is it an attempt to make the invisible visible in an artistic project with the aid of visual representations. His goal is to use the notion of hertzian space »as a catalyst, encouraging the poetic and multi-layered coupling of electromagnetic and material elements to produce new levels of cultural complexity.«

Cultural innovation is seldom mentioned in the discourse of technological innovation. The concept of innovation very often stays on a level of contextual use, providing improved functionality or more time at task in a very utilitarian way. The aesthetic experience in art relies heavily on the acts of interpretation and perception. Where recognition is used, a shift of meaning is probably involved. The shift of meaning is often an imaginative alternative to the actual. Infusing doubt about current situatedness puts the experiencing actor (he is not a consumer, referring to art as consumer goods would offend most artists I know if they don't wish to be very provocative to artist society and therefore view art as a consumer product) in a very active role. Interaction designers could well let digital media, systems or objects mediate this aesthetic experience. Dunne puts forth the idea of the user as a protagonist where use of objects makes a narrative context. In these psycho-social narratives the designer becomes very much an author by providing new behavioural opportunities without enforcing them. The way she does that is by presenting these opportunities, not by representing.

...the designer becomes very much an author by providing new behavioural opportunities without enforcing them.

A way to challenge the embodied meaning in artefacts is to deliberately misuse them and make up new scenarios distorting the original contexts of use. Dunne mentions perverts, using hand held scanners to scan body parts wrapped in plastics and then sending the copies to lonely women, as an example. A subtler example could be the way artist Krzysztof Wodiczko used monumental statues or buildings as surface for large-scale projections. By projecting for example a gun on an obelisque, which has the resembling shape of a gun and is placed in front of the ministry of defence, he temporarily changes the urban space as well as our conceptual understanding of it. In this way the user turns into an abuser and the notion of design becomes an inscription, a tactile commentary on ideologies instead of conservator of meaning. Wodiczko is originally an industrial designer and while video is a common material among artist's he has produced designed objects using electronic component's and digital media such as the Alien mouthpiece.

Latour (1999) describes our use of technology as a collective of humans and non-humans instead of the traditional subject/object ontology. The way he goes is by defining technology (here Latour uses the term techniques) as a mediator, not a

means and not an end but both at the same time. He gives a multi-faceted concept of technical mediation in where the term society is substituted by the collective, an exchange of human (»user«) and non-human (»object«) properties inside a corporate body. Translation is the term he uses for describing a drift or mediation in our intentionality while using technology. When an agent (a human, a user) gets hold of a tool in order to achieve a certain goal, the agent gets modified by the potential capacity of the tool. So what happens is that a slight shift emerges in the identification of not only agent and tool, but in the goal as well. The non-human (the technology) can have a goal as well in terms of it's intended function. Since the non-human has an equal impact on the translation Latour uses the term actant to describe both the human agent and the non-human technology. When a human falls back on technology in order to achieve a certain goal he makes a *detour* according to Latour. The intimate coupling of actants in this situation gives rise to a third actant, a combination of the two and as mentioned the goal is translated as well. Thus the action becomes a series of associated entities and nested sub-programs.

What artist Fischli and Weiss did was taking an imaginative approach to the detour and the collective. Design of the detour could dwell on the complexity and the potential of the meeting. I don't think there will be an optimal fit.

DANGEROUS FRUITS

Every game has two pre-conditions; the rules of playing and the playing with the rules. In natural play we can observe how children incorporate breaking of the rules into new ways of playing a game. Latours use of the term detour has a subversive sibling in the artistic avant-garde movement of the French Situationists, in a method mentioned as *détournement*. Détournement is the reuse of pre-existing elements in a new ensemble. It might be seen as negation of previous organizations of expressions and at the same time the organization of a new meaningful context. Whereas the Situationists have used it as an artistic-political weapon, it's also quite common in the language of commercial advertisements. The idea is also closely related to that of *bricolage*.

An example could be *The City Joker* project realized by Austrian architect Dieter Spath in Graz a couple of years ago. He and his colleagues simply drew a straight line on the city map across the city and decided to travel the line with the aid of the inhabitants living on the line.

A map is the city as a circulated referent to speak Latourian, and of course a straight line on a map crosses buildings, canals, churches, walls and so on. The project turned into a spectacle engaging the whole town but especially the people

living on the line. When the line crossed actual homes they made interviews with people and produced postcards portraying inner feelings of the inhabitants. The postcards where sent along the line. What happened was an emergence of a new neighbourhood in the city and people living on the line proudly referred to themselves as »line people« and kept staying in touch after the end of the project. It was a re-programming of the urban plans and an unconventional use of the environmental knowledge learned at architectural academies. It could be discussed whether such an act is useful or on how valid the scientific grounds beneath are. Never the less I think this re-programming capacity is something that is inherent in interaction design and I also see it as an extension of participation when handed over to users. Also use can't be pre-articulated.

Consider this naïve analogy of fruit from the Snowhite fairytale: The witch is »poisoning« the apples, Snowhite, also being naïve, is »eating« them. But if she pulls on a hood and throws the apples on a hunter passing by, in an attempt to reclaim the forest, she is »stoning« the apples. Clearly this would be a borderline case in the use of apples, but an important one, in that it informs us about the state of eating fruits. Use is more about verbs than nouns.

Para-functionality is the term Anthony Dunne uses for describing a diverse functionality that is within the scope of utility but overrides conventional functionalism. Some of the examples he gives are old and eccentric design ideas such as; »Walking sticks that become a card table or a seat show how simple props can transform architectural space. They conceptually colonise the functional possibilities of pre-existing space. The user becomes a protagonist in a new narrative where a lobby or park becomes a casino« (Dunne, 1999, p. 45). So by throwing these kinds of props into the everyday environment people can participate in stories and change conceptions of reality. It's also an act of configuring and reclaiming space as opposed to pre-formatted space/environment. Again it's an argument for exploring boundaries of the actual and the imaginative by the use of designed objects. Other examples are from art or marginalized or obscure electronic objects like tools for surveillance and scanning radio frequencies.

I think it's important that these ideas of design as critique or tactile commentary not stay on a conceptual level, reside in an art gallery or are articulated as witty jokes or spectacles.

Here I disagree with Dunne who prefers the conceptual model to the working prototype.

We need a discourse and good texts. But also design ideas are ideally articulated in implemented design objects if these kinds of ideas don't want to be seen as a fringe phenomenon. To use Latour's perspective of detour would mean that the

actual meeting between user and object resides not in a conceptual model but in actual use.

Neither would it be fruitful to invent a new dichotomy such as that of functional/surreal. The rational/irrational combat field is old and not promising. Design could well work from within function with implemented articulations of these ideas, but with an extended notion of »utility.« Hallnäs and Redström (2002) bring forth the concepts of *presence* and *expression* for describing to what degree we accept and invite artefacts as meaningful building blocks into our lives. This accepting of computational things is not something that takes place in a distinct time slot for fast and efficient interaction, »...when thinking about the interaction in terms of dwelling, the time spans in focus are much longer, for example days, weeks or even years« (ibid., p. 117). This consciousness of temporal gestalt is different than just acknowledging that an artefact will be used for a long time. The notion of utility and that of presence must not be treated separately but are different perspectives that must be seen as complements in design of digital artefacts.

Design as critique should also just not be an interpretation of visual codes and fantasies of future scenarios, it must be considered in a context of actual use.

The game is about elevating the identity of the interaction designer from just an executioner of utility to that of an independent architect of use with artistic and theoretical ambitions of his own. Clearly this is not the market right now, but consider the role of the architect who is well respected for infusing theoretical programmes into projects that are actually built and used parts of urban plans.

INHABITATION OF INTERACTION

I am really fond of Donald Norman and human factors perspectives have been very useful to designers. But maybe they are more suited for contexts of products and work practice. Also they are more focused on interface architecture than contextual experience. The way not only media and tools go computational, but also our surrounding environment, down to the very fact that even matter is programmable, calls for awareness of use that is grounded not only in terms of utility. The emergence of ubiquitous or pervasive computing and responsive technologies increases complexity in issues of use. The specific task just isn't around anymore. When it comes to smart materials we also lose the concepts of absolute form and natural properties. Mechatronics, sensors and tags are making up new environments built upon networks of semi-autonomous objects where information will be spatially configurable. We can continue to talk about these environments as embedded systems with the hazard of just continuing the engineering role of the interaction

designer, but I would prefer to talk about a new kind of space in where interaction design teams up with architecture. It will be a great challenge to keep these systems compatible and open for integration, but we'll also have to extend our notion of aesthetics.

Architecture is a field where critique of functionalism is old. The blowing up of the Pruitt Igoe complex was a physical articulation of not wanting a hostile environment for living. Functionalism was attacked on theoretical levels from neo-modernists and by post-modernists for aesthetic reasons. Modernist theories in literature, art and film also formed a base for some architects and »form follows form« replaced »form follows function« (Tschumi, 1994, p. 116). There's an interesting fact underlying this internal referencing. Architecture had seen that function survived through exchanging contexts. A fast-food restaurant becomes a pawn-shop and libraries turns into swimming pools. Rem Koolhaas described an Athletic Club as »Eating oysters with boxing gloves naked on the ninth floor«(Tschumi, 1994, p. 256). To this rapid exchange of movements in space we must add a diversity of cultural, economical and individual perspectives. So for architects the »stoning« of the hunter is not a possibility, it's a natural condition. The old Vitruvian trilogy of venustas-firmitas-utilitas is opposed to a view where design conditions are in constant change. These disjointed conditions were the topics of many post-modern philosophers in the 80s. Architect Bernard Tschumi was one of several that fetched inspiration and methods in that movement and maybe especially from Derrida and his concept of deconstruction. Instead of excluding functionality, which in his terminology is called event or programme, he has stated that there are no hierarchical cause-effect relationship between form and function and that that very condition is the strength of architecture. He defines program as »a list of required utilities, it indicates their relation but suggests neither their combination nor their proportion« (Tschumi, 1994, p.113). One could argue that this is the way that the human factors tradition design software or systems. But, instead of trying to ease the gap between user and artefact, Tschumi like Dunne uses the gap as inherent in design.

He goes very explicit about it, using a metaphor of architecture as violence, mediating a shock. So instead of providing continuity in urban spaces in the old form of streets and plazas he sees fragmentation and juxtaposition in cities like New York and Tokyo as a sign of vitality, making an event out of urban shock (ibid., p. 246). Superimposition, juxtaposition and de-familiarization have become key devices. For Tschumi architecture is both concept and experience, space and use, and they are never separated. Instead they are merged into unprecedented combinations of programs and spaces. He calls this crossprogramming. An example would be to take the spatial configuration of a prison and combine it with that of an event

park. These combining of events is not limited to a series of actions but is meant to act as a problemization of a situation of use as to expand into a wholly new setting. Derrida also suggested to him that the word »event« shared roots with that of »invention.«

CONCLUSIONS

If we want an identity of the interaction designer beyond engineering perspectives we have to extend definitions of usability. This is especially necessary for the emergence of new kinds of spaces where interaction design becomes an environmental factor in social spheres outside of workspaces. Some of the traditional HCI perspectives might be contradictive to innovation, especially cultural innovation. Aesthetics of use is equally important as ease of use. Design must also support users taking part of a social narrative. This doesn't imply to throw all valuable knowledge acquired from the HCI tradition overboard. Design could well work from within function, but with an extended notion of »utility.« These ideas I think should strive for implementation and not stay put on a conceptual level. Very roughly some steps in the shift from HCI to interaction design could consist of viewing digital systems not as a tool for instrumentality but as a mediator of experience. Digital systems and objects go through a series of transformations and unspecified contexts of use in a complex chain of »Latourian« detours. It will be a great challenge to design these detours without thinking of use situations as fixed but as dynamic and process oriented. If the price is relying on users' interpretation rather than understanding in our mediation of technology, so be it. I think they will.

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